Brazil's Missing Infants: Zika risk changes reproductive behavior*

MARCOS A. RANGEL
Duke University
and BREAD

JENNA NOBLES
University of Wisconsin, Madison

AMAR HAMOUDI
University of Wisconsin, Madison

First version: *Apr 2017*
This version: *Mar 2019*

Abstract

Zika virus epidemics have potential large-scale population effects. Controlled studies of mice and non-human primates indicate Zika effects on fertility, raising concerns about miscarriage in human populations. In regions of Brazil, Zika risk peaked months before residents learned about the epidemic. This spatio-temporal variation supports differentiation between Zika biological effects on fertility and learning about its risks over reproductive behavior. Causal inference techniques used with vital statistics indicate that the epidemic caused 20% reductions in birth cohort size 18 months after Zika infection peaked, but 10 months after public health messages advocated childbearing delay in the most affected areas. The evidence is not consistent with biological reductions in fertility being the main force; it indicates strategic changes in reproductive behavior to temporally align childbearing with reduced risk to infant health within locations where Zika risk was more salient to the population. The effects are larger for the more educated, older and wealthier mothers, which may reflect their facilitated access both to information and family planning services within the high-risk/mosquito-infested urban locations.

JEL:J13, O15, I15, I18, I12

Keywords: Zika, fecundity, fertility, avoidance, *tempo* effects

*The authors thank Elizabeth Frankenberg, Sarah Hayford, Malia Jones, Giovanna Merli, Seth Sanders, Duncan Thomas, Jenny Trinitapoli and PAA 2017 Meeting participants for comments and suggestions. Funding support was provided by NICHD (1R03HD092818-01), the University of Wisconsin-Madison Graduate School, the Center for Demography and Ecology at Wisconsin (NICHD P2C HD047873), and the Sanford School of Public Policys Pilot Project Fund. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

The Zika virus epidemic has spread rapidly across the Americas, with recent emergence in South Asia, raising a number of concerns about its impacts on population welfare. Such impacts include threats to maternal and infant health, as well as to both male and female reproductive health and viability. In this study, we use multiple data sources from Brazil to provide evidence on the effects of the epidemic on population fertility. We employ a study design developed for causal inference that distinguishes the epidemic's biological and behavioral effects and considers contemporaneous political and economic upheaval in Brazil. The design allows us to expand on the observation that fertility rates have fallen in affected nations. The research produces two important results: i) the fecundity effects documented in controlled animal studies have not manifested into large drops in births in human populations; ii) there is evidence of a remarkable and relatively immediate behavioral response to information about the epidemic, and; iii) behavioral response is differential across education and age groups, corresponding to significant changes not only on the size, but also on the composition of birth cohorts.

Demographic responses to new health risks provide a window into a central component of fertility: whether, and when, people adjust reproductive behavior to time births with favorable environmental conditions. Most models of fertility suggest behavioral links between reproduction and contextual conditions like macroeconomic change, epidemiologic threats, agricultural yield, or war. In these models, fertility preferences and behaviors vary across the life course, and may respond to shifting circumstances, including the costs of investing in children, the desirability of being pregnant, and changes in expectations about the future (Becker and Lewis 1973; Hayford 2009, Hotz, Klerman, and Willis 1997, Jones et al. 2015, Johnson-Hanks et al. 2011, Kim and Prskawetz 2010, Preston 1978, Schultz 1997). Some of these changes manifest in increases or decreases in the total number of children born (quantum effects), but they may also manifest in temporal shifting of births (tempo effects), so that births are timed to occur when parents are better positioned to bear and raise a child (Schultz 2001; Jones 2014).

Empirically, fertility has been observed to decline during periods of hardship like conflict, disease exposure, and economic decline (Agadjanian and Prata 2002, Lindstrom and Berhanu 1999, Sobotka et al. 2011, Terceira et al. 2003), and to increase following the eradication of some forms of disease, like malaria (Lucas 2013). Nevertheless, interpreting the alignment of demographic rates and contextual health threats as evidence of strategic change in reproductive behavior is difficult (Alam and Portner 2016, Barreca et al. 2018, Hill 2004, Kim and Prskawetz 2010). In many cases, the factors that increase the costs and/or risks of childbearing also affect other proximate determinants of fertility. Disaster, conflict, or epidemic may result in family separation and an accompanying reduction in coital frequency. Health threats may also reduce biological fecundity (e.g., Barreca et al. 2018, Terceira et al. 2003). Changes in exposure to nutrition, infection, and multiple forms of physiological stress have effects on organ systems centrally involved in establishing a pregnancy and carrying it to term (Arck et al. 2008; Larsen et al., 2013). In addition to these alternative explanations, the standard threats to interpretation in observational research are also relevant. Other coterminous phenomena that exacerbate disease risk—like political or economic instability—may simultaneously shape fertility through biological and behavioral pathways.

The recent observation of fertility reductions in populations affected by Zika are similarly difficult to interpret (Coelho et al., 2015; Castro et. al., 2018). Though Zika may have both biological and behavioral effects, the biological effects on fertility have received far more attention in scientific research and the popular press (e.g., Beaubein 2017). Zika has well-documented effects on fecundity in mice and, to a lesser extent, non-human primates (Dudley et al., 2018, Govero et al., 2016). As a result, several scholars have questioned whether Zika contributes to population-level reductions in fecundity, raising concerns about the global implications of the epidemic (Coelho et al., 2015, Esposito et al., 2017, Gonce et al., 2018).

At the same time, fertility effects via changes in reproductive behavior are plausible. The health risks associated with Zika were elevated in public consciousness through widespread media depictions of infants with microcephaly. These were accompanied by unequivocal public health recommendations to delay fertility. Qualitative evidence from Northeast Brazil describes women with clear knowledge of Zika's health risks (Borges et al., 2018), and for some, a desire to delay childbearing (Marteleto et al., 2017).

It is also, of course, possible that population fertility declines could be entirely unrelated to Zika risk. The last five years in Brazil have been marked by significant political and economic unrest, including the impeachment of the nation's president and a 2014/2015 reversal of the nation's previously strong GDP growth. As we demonstrate in this study, fertility in Brazil has closely followed economic conditions for many years. Accounting for these coterminous effects is necessary to ascribe population-level fertility change to any factor related to the epidemic.

To distinguish these interpretations, we leverage the spatiotemporal spread of Zika in Brazil. We pay particular attention to the period when residents in Northeast Brazil were at risk of infection but had not yet learned about Zika. We contrast these plausibly biological effects with the changes in fertility observed after information about Zika was widely distributed. We use two quasi-experimental design approaches to rule out competing explanations for behavior.

We draw on multiple data sources, including complete natality files from the Brazilian Birth Registry, data on the geospatial penetration of the primary mosquito vector (*Aedes aegypti*) for Zika spread, data from hospital accounting records documenting regional counts in procedure codes (*Sistema Unico de Saude*), as well as Census population data, information on labor markets and inflation, and United Nations data on regional economic and infrastructure development.

By unpacking Brazilian demographic change accompanying the Zika epidemic, our analysis sheds light on a central aspect of fertility: whether, when, and which people time births to align with reduced risks to maternal and infant health. Behavioral timing effects are widely theorized but not well-documented empirically. The results also demonstrate evidence of public response to explicit anti-natalist policy recommendations from the federal administration. This particular case is all the more striking because realized fertility reductions occurred in a region of Brazil in which the majority of births are unintended and abortion is illegal and difficult to access. This is also highlighted when we uncover that subsections of the population were more responsive than others. In particular more educated, wealthier and older women were the first and most likely to reduce fertility.[1] Before turning to the study design, we describe the Brazilian context and the spread of

[1] Compositional effects are important for the evaluation of the long-term effects of the epidemics in the most affected localities, like the case of curbing access to certain family planning methods discussed in Pop-Eleches (2006).

Zika risk in more detail.

ZIKA VIRUS EPIDEMIC IN BRAZIL

The onset of a public health emergency

The Zika virus was originally identified and named in 1947 after macaques with fevers were examined in a small forest in Uganda. For at least sixty years, the virus received very little global attention from scientists and public health officials, until human infections were detected in Micronesia (2007) and French Polynesia (2013).[2] It was not until April 2015, when the Brazilian Ministry of Health identified a Zika outbreak in the northeastern region of the country and notified the World Health Organization, that the virus gained global notoriety.[3] Over the following year, the outbreak intensified and spread, reaching into the southern United States and more than forty other countries in the Americas. By mid-2017, the number of Zika cases in the region exceeded half a million (PAHO, 2017). Genetic findings reported by Faria et al. (2016) indicate that the virus most likely arrived in Brazil during an international soccer tournament in June 2013, which had host cities located in areas known for their suitability to the *Aedes* mosquitoes which transmit the virus (Weaver et al., 2016; Messina et al., 2016), and observed high traffic of visitors and presence of tourists from French Polynesia.[4]

The arrival of Zika in that region of Brazil made its clinical detection difficult to distinguish from the more common Dengue and Chikungunya fevers, which have seasonally afflicted urban populations across Brazil for more than two decades. Up to four-fifths of Zika-infected adults are asymptomatic (Duffy et al., 2009) and to date, there is no indication that Zika affects prime-age adult mortality. The epidemic raised special concerns once evidence suggested that Zika increases the risk of severe neurological congenital disorders among infants and the risk of Guillain-Barré

[2]See McNeil Jr (2016) and the World Health Organization (n/d).

[3]Reports of a new disease appeared as early as the third trimester of 2014. Laboratory tests later confirmed cases in Pernambuco in December 2014, and in Maranhao, Rio Grade do Norte, and Bahia, in February and March 2015.

[4]Tahiti was one of the national teams participating in the Confederations Cup. One of their matches, in June 23 2013, was played in Recife, Pernambuco.

Syndrome among adults (Epelboin et al., 2017; Mkakar et al., 2016; Molnar and Kennedy, 2016; and De Oliveira et al, 2017). By late October 2015, Brazilian health authorities were informed of an unusual increase in the number of cases of microcephaly among newborns within the same region of the country. Because cranial growth accompanies brain increase, microcephaly indicates serious risk of developmental delay, including problems with speech, standing, walking and multiple intellectual disabilities.

On November 17, 2015 the detection of Zika virus in amniotic fluid samples from two pregnant women with microcephalic fetuses were confirmed and triggered the declaration of a national medical emergency by Brazilian health authorities.[5] The Ministry of Health encouraged pregnant women to take preventive action against disease-spreading mosquitos and women in family planning were advised to consult with doctors about risks to their fetuses. Cases of congenital complications started being tracked and reported weekly; Zika suspected and confirmed cases were tracked as well.

As the epidemic evolved, government agencies across the Americas advised pregnant women to avoid mosquito-infested areas.[6] Soon after the declaration of medical emergency, officials in the Brazilian Ministry of Health took the unprecedented step of advising women to consider delaying fertility indefinitely (McNeil Jr. 2016, Castro 2016).[7] At the time, the dramatic increase in positive cases and their very graphic display in pictures was reported regularly by the media (Ophir and Jamieson 2018, Ribeiro et al., 2018). We reproduce the timeline of these events overlaid with the dramatic increase in the incidence of microcephaly across the country in Figure 1 below. The number of microcephaly cases recorded per live birth in December 2015 is approximately 28 times larger than the norm since 2005. The 12-month aggregated rate peaks in the second half of 2016,

[5]Once mothers-to-be are infected, the risk to pregnancy outcomes is uncertain (Honein and Jamieson 2019); the most recent estimates of the risk of congenital anomalies range from 0.03 to 17% (Brasil et al., 2017; Jaenisch et al., 2017; Mulkey et al. 2019; Rice et al., 2018).

[6]The Centers of Disease Control and Prevention (CDC) has installed a Task Force on Pregnancy and Birth Defects in response to the Zika scare. The CDC has also increased its efforts to communicate the importance of the massive growth in mosquito-borne diseases in the United States over the last decade in an attempt to increase avoidance and prevention among the population.

[7]Zika was then added to the list of microcephaly-correlated factors, like severe malnutrition, substance abuse, infections with rubella or toxoplasmosis, and some conditions that interrupt blood supply to the fetus.

revealing a 10-12 fold increase in incidence relative to previous years.

[Figure 1 about here]

Not all geographic regions in Brazil are equally suited to support sustained autochthonous transmission of the Zika virus. *Aedes* mosquitoes, which are primarily responsible for transmitting the virus from person to person, reproduce most readily in warm humid regions with high levels of rainfall. Scholars have constructed geospatial Zika "suitability" indices based on detailed climate, topographic, and land use data (e.g., Messina, Kraemer, et al. 2016) alongside information about population density and mobility to model the spread of the disease (Zhang et al., 2017). This type of modeling is preferred to the use of Zika diagnosis counts to understand the spatiotemporal distribution of infection risk because diagnosis counts are widely understood to understimate risk. Diagnosis counts also skew the picture of the geographic/socioeconomic distribution of risk; undercounts are likely worse in lower-income relative to higher-income communities. This type of modeling, alongside information about microcephaly incidence, quickly elevated the microregion of Recife and other urban centers in the state of Pernambuco and across the Northeast as the centers of Zika risk in 2015 and early 2016 (PAHO, 2017).

Fertility impacts of a public health emergency

The Zika epidemic emerged in a period of ongoing fertility decline in Brazil. The nation's total fertility rate fell from 6 births per woman in 1960 to 2.1 by 2005 and 1.8 by 2014. Contraceptive use increased, though with considerable variation across the country. Data from the most recent Demographic and Health Survey (2006) indicated that 55% of pregnancies were unintended.

Whether a contextual change like the Zika epidemic affects fertility behavior depends in large part on (i) how its risks are perceived by the population and (ii) the resources available for adjusting behavior (Awkwara et al. 2003, Yeatman 2009). To date, research on the effects of disease risk on fertility has largely focused on disease eradication. Many of these studies rely on a framework in which parents have a targeted number of children and adjust fertility behavior to perceptions of child survival (Preston 1978, Boucekkine et al., 2009). Bleakley and Lange (2009) study the eradication of hookworm in the American South and observe associated decreases in fertility. They argue that because hookworm had minimal effects on adult health, fertility reductions resulted from an increase in the returns to investments made in children. Ager, Hanson, and Jensen (2017) observe reductions in the number of children ever born in Sweden following smallpox vaccination; they attribute these to reductions in child mortality. McCord, Conley, and Sachs (2017) similarly find positive effects of child mortality from malaria on population fertility. Lucas (2013), by contrast, observes an increase in fertility after malaria eradication in Sri Lanka, a change largely attributed to increases in pregnancy survival, versus effects on fertility behavior, specifically.

The primary example of a *new* epidemic threat in this literature is HIV/AIDS, about which much has been written. A distinguishing aspect of HIV is its effects on adult mortality. Several studies suggest that some families seek to initiate births earlier, in part because of concern about future risk of HIV infection and associated lifespan uncertainty (see Nattabi et al. 2009, Trinitapoli and Yeatman 2011, Yeatman 2009, Hayford, Agadjanian, and Luz 2012). Other studies similarly identify population level fertility increases associated with HIV risk (Chin and Wilson 2018), though the general conclusions about population-level associations in this line of work appear sensitive to identification (Kalemli-Ozcan 2012).

A general insight from the disease risk literature is that fertility effects depend on risk perception. For example, perceptions of risk are amplified by experiences within individuals' social networks and their local communities; spatial clustering of disease risk may have an outsized effect on behavior. Institutions with political authority may exert stronger effects on risk perception. And perceptions may respond asymmetrically to increases and decreases in true health risk; risk increase may more acutely observed than risk reduction (Montgomery 2000, Kohler et al. 2007). As a result, information media—both formal reporting and social media—likely impact the construction of risk perception.[8]

[8] Social media are becoming an increasingly important conduit of information about health risks (Fung et al. 2014; Panagiotopoulus et al. 2016). Media representations often inflate infectious illness threats relative to their contribution to population morbidity and mortality (Frost et al. 1997, McComas 2006).

Each of these appears relevant to population perception of risk in the case of the Zika epidemic. In Brazil, information about Zika has been distributed through official government bulletins, professional journalism, and on social media. During February 2016, over 200,000 members of the armed forces went door-to-door in a targeted attempt to distribute information and disrupt mosquito breeding grounds (Portal Brasil 2016). While that action was warranted for vector control, the presence of the armed forces may have raised more concerns regarding the prevalence of mosquito vectors, than assured population confidence that the epidemics could be controlled. Such interactions from the formal authority of military and public health institutions provided relatively transparent information about area-level Zika risk. By contrast, variation in individual-level risk may have been more difficult for residents to assess.

For women and couples in Brazil, integrating this new threat into fertility plans may have been challenging (Simmons and Rigby, 2016; Marteleto et al., 2017). Several of the regions more suitable for Zika transmission are also the least resourced parts of the country. Delaying fertility indefinitely may not be an option for women and families, especially where access to contraception is limited and childbearing is a central feature of family life. Moreover, several of Zika virus' hypothesized effects cannot be discerned via ultrasound until the second and third trimesters (Oliveira Melo et al., 2016), which may limit the ability to legally interrupt pregnancies or to do it without elevated risks to maternal health.

The capacity to react to these risks almost certainly varies *within* the population. In 2012, contraceptive use among women 15-49 was close to 75% in the wealthier Southern states and only 30% in Pernambuco (Minowa et al. 2015). Abortion is illegal and the access to clandestine alternatives vary by socioeconomic conditions, which are heterogeneous even within well-defined urban microregions like those examined in this study.[9] Well-resourced areas in the Southeast (and households within any locality) are more connected to up-to-date information about the epidemic and have higher density of social-media networks, even though risk of local infection may be lower

[9] A recent exception, introduced by a Supreme Court ruling in 2012, involves the abortion of anencephalic fetuses. There is widespread debate over what this ruling implies for other congenital anomalies but the current legal reasoning seems to indicate that detection of microcephaly would not offer grounds for a legal abortion.

amongst them. Ultimately, a differential temporal change in behaviors depends on the salience of both the risk perceptions of individuals *and* the ability to react to those.

In Brazil, a monthly count of live births indicates a fall of up to 15% in the months following the outbreak compared to the months preceding it (Figure 2). In this study, we seek evidence that these declines represent a causal impact of the epidemic. We then look to move beyond other research on population level fertility effects of disease by distinguishing whether these represent evidence of behavioral responses to the disaster or, as others have argued, these represent concerning information about the epidemic on human fecundity (Coelho et al. 2015).

A primary issue in identifying the effect of Zika exposure over fertility in Brazil is the potential effect of contemporaneous economic and political instability. Starting in the second quarter of 2014 the Brazilian economy initiated a downward spiral which by the end of 2016 had become the worst recession since 1981. Between 2014 and 2016 the gross domestic product (GDP) for the country *fell* 7.2%, being the first time in 70 years the country observed two years of back-to-back contraction of the economy. Despite some improvement during 2017, by the end of that year the Brazilian economy was still at the same level it was in 2012. From the first steps towards a contraction in 2014, the overall expectation of economic agents already reflected the turbulent conditions that would follow, particularly after the first half of 2015.[10] Figure 3 overlays the trends for year-to-year changes in 12-month birth cohorts (which approximate 6% by the end of 2016) with the year-to-year changes in consumer prices and unemployment rates. It is clear that any study of Zika effects over population change must be attentive to the potential impacts of broader macroeconomic conditions (and expectations) over fertility decisions.[11]

[Figures 2 and 3 about here]

Political instability also affected the country during the time Zika came into the picture. From

[10] http://g1.globo.com/economia/noticia/2014/12/o-que-esperar-da-economia-em-2015.html; http://g1.globo.com/economia/noticia/2015/12/o-que-esperar-para-economia-em-2016.html

[11] A large literature spanning the social sciences suggests that fertility co-moves with the economic business cycles, with improvements on the latter generating increased fertility (see Adsera (2004, 2011), for example. A recent contribution by Buckles et al. (2018) has even uncovered evidence that fertility decisions are a leading economic indicator, with conceptions moving in anticipation of bad and good future economic tides.

December 2015 to April 2016, President Dilma Rouseff was under a Congress-led investigation which led to an impeachment halfway through her second-term in office. Her vice-president and successor, with little popular and political support, faced challenges to govern and did very little in multiple public policy fronts. One can even argue that the combination of economic and political crises may have contributed to an overall belief that health shocks like Zika could not be combated by a failing public sector apparatus, increasing in this way the perception of risk during the epidemics.

For all these reasons, methods that generate expected period changes in fertility in the absence of Zika exposure need to resemble as closely as possible a case-control exercise. Castro et al. (2018), for example, employ Auto-Regressive Integrated Moving Average (ARIMA) time-series forecasting methods to generate expected fertility rates in Brazil based on past trends. This is a useful strategy to demonstrate temporal deviation from the past, but does not provide leverage to interpret trends if other factors that affect fertility are changing concomitantly with the exposure of interest. In order to advance our understanding we must attend to the macroeconomic and political turbulence of the time. Below we delineate inference strategies that leverage spatiotemporal variations in order quantify changes in fertility and a potential behavioral explanation for the observed decline: health-risk avoidance on the part of some would-be parents.

DATA AND METHODS

Data

To investigate fertility patterns in the wake of the Zika epidemics we use information on all births in Brazil between January 2011 and December 2017 from the Brazilian Birth Registry. The registry has high coverage rates, exceeding 98% as early as the mid-2000s (Mello-Jorge et al., 2007). These records include information on mothers' age and educational attainment, municipality of maternal residence and municipality of delivery. They also record birth outcomes, including the diagnosis of any congenital anomaly (coded based on the World Health Organization's International Classification of Diseases, ICD-10). These vital records are public and posted on the Ministry of Health web portal.[12] As of this study's writing, records do not exist beyond December 2017 (with all records for 2017 labeled as preliminary). The registry includes microdata on every live birth across 5,570 municipalities, the smallest administrative unit in Brazil.

We aggregate this information to the generate monthly birth counts by microregion (or *Região Geográfica Imediata*). Microregions are defined as areas containing common employment and goods markets; they also provide catchment areas for health and education services. We use the boundaries published by the Brazilian Statistical Institute (*Instituto Brasileiro de Geografia e Estatistica, IBGE*) in 2017. The birth registry data contain information for all 510 microregions (see map in Figure A.1 in the Supplementary Online material). To ensure that the monthly estimates are not driven by noise from sparsely populated regions, we restrict the working sample by trimming the most rural areas: microregions that between 2011 and 2013 had at least one month with live-birth counts in the bottom 10% of the microregion-month live births counts' distribution. The resulting sample includes 412 unique microregions, which together represent 96.3% of all births for residents in the Brazilian territory during that period.

We use several data sources to characterize socioeconomic and demographic variation across the regions. We measure variation in poverty and economic development with the United Nations human development index (HDI), constructed for each municipality in 2010. To do so, we aggregate the municipal HDI data to the microregion level using 2010 population counts as weights. We also draw on three subcomponents of the UN HDI index which classify regional variation in income, health, and educational attainment, respectively (UN n/d). We compute urbanization rates, population size and population age composition using microdata from IBGE's 2010 census of population. To describe the evolution of macroeconomic conditions in the country, we employ information regarding labor markets and inflation made available by the *Instituto de Pesquisas em Economia Aplicada* (IPEA) – IPEADATA.[13]

[12] *Sistema de Informacao de Nascidos Vivos*, (SINASC-DATASUS), for live births; and *Sistema de Informacao de Mortalidade*, (SIM-DATASUS), for registry of fetal deaths.

[13] http://www.ipeadata.gov.br/Default.aspx

Because this study is centered around a health emergency associated with a mosquito-transmitted disease, we retrieve reports published by the Ministry of Health's *Secretaria de Vigilancia em Saude* on the Larval Index Rapid Assessment for *Aedes aegypti* (*Levantamento de Indice Rapido do Aedes aegypti*, LIRAa). The index is constructed from a survey of localities in which researchers inspect for mosquito eggs in standing water through door-to-door visits. The eggs found are collected and tested in a lab for the presence of *Aedes aegypti* larvae. The results are used to construct an index indicating the percentage of households visited with positive tests for *Aedes* mosquitos (*Indice de Infestacao Predial*, which heretofore we refer to as LIRAa Index). This index is reported widely in Brazil. Locations with more than 3.9% of their households testing positive are considered high-risk (or red-flagged). The communication of these findings is central to combating mosquito-borne diseases: incentivizing the population to take action by eliminating standing-water within their households, and guiding local authorities in organizing the spraying of streets with insecticide.

In supporting analysis, we also use data on hospitalizations for institutions (both public and private) which are affiliated service providers for the *Sistema Unico de Saude* ("Unified Health System" or SUS). The SUS system provides routine medical services for approximately three-fourths of the population in Brazil with the rest occurring in private clinics unaffiliated with SUS.[14] To reimburse providers, SUS requires the maintenance and transmission of records on all activities, including date of visits, procedures, diagnoses, length of inpatient stays, and basic demographics on patients.[15] Both municipality of service provision and patient municipality of residence are recorded. These data are reported in public databases with a two-month delay and go under recurrent updating during a 12-month moving window. Our experience is that these data are nearly complete at the six- to eight-month lag (http://www2.datasus.gov.br/DATASUS/). Diagnoses are reported following ICD-10 classification.[16]

[14] Despite the high coverage, we expect more-educated, high-income and privately insured individuals to be underrepresented within the SUS's hospitalization records.

[15] Collected under the *Sistema de Informacoes Hospitalares*, (SIH-DATASUS).

[16] This includes codes for vaginal and c-section deliveries (O80- Encounter for full-term uncomplicated delivery and O82- Encounter for cesarean delivery without indication, and deliveries with complications under code O60-O69). Unfortunately for our purposes, the data provide little information on contraception because most methods do

Methods

The nature and timing of the Zika virus epidemic in Brazil provides unusual leverage to distinguish behavioral fertility responses. In some locations infection risk peaked before most Brazilians knew what Zika disease was or its potential impact on infant and maternal health. Therefore, if population fertility were to fall because of reductions in fecundity, we would expect missing births to be first noticeable roughly 8-10 months after the outbreak had begun, corresponding to the gestation period of most successful pregnancies. If spontaneous pregnancy loss contributed to fertility decline, missing pregnancies may be detectable a few months earlier; i.e., 6-8 months after the outbreak began. By contrast, fertility decline resulting from behavioral attempts to delay fertility and avoid pregnancies (including elective termination) would be expected at a later time period, at least 6-10 months after the announcement of the epidemic and the peak in microcephaly cases.

On top of these temporal patterns we also expect the spatial distribution of suspected Zika cases to influence fertility patterns. Biological effects should be clustered in regions in which infection risk was highest. Behavioral changes should also be more likely in locations in which the epidemic was perceived to pose a greater threat—regions in which microcephaly cases were more numerous, and where insecticide campaigns and information distribution were more intense. Again, these behavioral effects should emerge later.

To identify biological and behavioral effects on fertilty, we use two complementary strategies. Both are methods that generate expected period changes in fertility in the absence of Zika exposure which resembles as closely as possible a case-control exercise using observational data. The first focuses on effects detectable in the metropolitan region of Recife in Northeastern Brazil. Recife was unambiguously the epicenter of the Zika-related microcephaly epidemic. Figure 4 contrasts the location-specific average microcephaly rates in Recife reproduce with other microregions in Pernambuco (17 microregions), across the Northeast (113 microregions) and elsewhere (281 microregions).[17] Recife was also a region in which the infection risk peaked in March 2015 (Zhang

not require hospitalization (excepting tubal ligation and vasectomies)

[17] See Figure A.2 for a a map of the Northeast region and of its microregional subdivisions.

et al., 2017), months before residents knew about Zika or learned about its risks. To isolate the epidemic effects in Recife, we employ a conceptual extension of standard comparative case control designs known as the synthetic control method (Abadie et al., 2010; Abadie et al., 2014; Doudchenko and Imbens, 2016; Arkhangelsky et al., 2019). This approach selects comparison series for Recife by combining information from other microregions that the analyst selects as potential "donors". The method is data-driven, and weights a combination of the comparison regions to find a best-fitting "synthetic" control during the period *prior* to the event of interest.

[Figure 4 about here]

We then complement the analysis with a second strategy, a more traditional difference-in-difference approach, that compares the fertility trajectories in exposed regions to the trajectories we would have expected, had fertility in exposed regions followed the path observed in unexposed regions. The approach also allows us to extend the analysis beyond Recife to several of the other affected microregions in the Northeast region of Brazil. We consider that both the spread of the disease and its salience to the population was a function of mosquito infestation—because of evidence that infestation was highly correlated with microcephaly rates and drove targeted public health measures. We use the mosquito surveillance indices published in January-February 2015 and October-November 2015 to characterize microregions in which we expect larger and earlier incidence of Zika infections and, consequently, any behavioral response to it. We describe each approach in more detail, below.

a) Synthetic Control

Recife, the capital city of Pernambuco, Brazil experienced the earliest clustering of Zika risk and received the largest share of media attention to microcephaly cases when the outbreak was recognized and announced. To use the synthetic control method, we construct time series of monthly crude birth rates (CBR) for Recife and all potential comparison microregions. Monthly crude birth rates are computed by dividing counts of live births by total population (in thousands) reported in the 2010 census.[18] In the synthetic control design, the analysts select a set of potential controls based on observed characteristics. Time series from these controls are used to predict the time series of interest (here, the CBR) in the exposed region during a period prior to exposure. Cases with low prediction error during the prediction period are weighted to create a combined synthetic control series. The vector of weights is generated by minimization of overall mean squared prediction error (Abadie et al., 2010; Abadie et al., 2014; Doudchenko and Imbens, 2016).[19]

We chose potential comparison regions that (i) were exposed to Zika later than Recife and with a much lower infection rate and (ii) may be similar to Recife on major characteristics that influence fertility. For this reason, we selected heavily urbanized microregions[20] located outside of the Northwest region of Brazil. These decisions produced 187 microregions in the pool of possible controls. We then used several attributes for the matching process, in addition the CBR time series. These included (a) the municipal level human development index - income sub-index; (b) human development index - health sub-index; (c) human development index - education sub-index; (d) the 2010 share of population in urban sectors; (e) the 2010 population share of women in seven distinct birth cohorts; (f) the 2010 share of births to women 35 or older, 30 to 34, 25 to 29 and under 24 years of age; (g) the 2010 share of births: with very-low (<1500 grams), with low birth weight (<2500 grams) and deemed premature; (h) the 2010 share of births women with at least high-school education, with primary but incomplete high-school education, and with less than elementary education; (i) the 2010 share of births by race and gender; (j) the 2010 share of babies born in private health units; (k) the 2010 rate of incidence for microcephaly and other congenital anomalies; (l) the 2010 share of population in urban sectors, and; (m) the 2010 local Gini coefficient for income distribution. In Table A.1 in the Supplementary Online Materials we present the descriptive statistics on these dimensions for Recife (column [1]) as well as for the

[18] We use CBR values, versus total fertility rates (TFR), and explicitly adjust for variation in the population age composition.

[19] There is some discretion on the implementation of the method, including which attributes to use in the prediction and which period to use in order to fit the time series of interest. We make our choices clear in this section and examine robustness of findings with respect to those when discussing robustness of results.

[20] We use as a cutoff for urbanization rates, measured from the 2010 census data, the rate for the least urbanized of the state capitals: approximately 77%.

chosen pool of microregions used to construct the synthetic contrast (column [2]).

We fit the data during the period up to March 2014, which is one year prior to the peak of Zika infections in the Northeastern portion of the country. We selected this window so that the calibration of the model occurs during a period that precedes the spread of the disease and related information. In Table A.2 (Online materials), we report the list of locations and respective weights that form the synthetic control series. Descriptive statistics are presented in Table 1, column 3.

Figure 5 presents three crude birth rate series (in log scale): for Recife, for the average non-Northeast urban microregion, and for the "synthetic-Recife" constructed using the method. Vertical lines indicate timing of major epidemic-related events and the cutoff for the fitting of the synthetic control. The simple average of the comparison microregions consistently underestimates the CBR level for Recife and displays month-to-month movement at odds with Recife, including during the pre-event period. By contrast, the series for synthetic-Recife is constructed to produce the best match of levels and seasonal variation in the period prior to March 2014. The weights used to match during the pre-March 2014 period are applied to the data following March 2014 to provide a plausible counterfactual for the time trend in fertility in places with low to no Zika risk in 2015 to 2017.

[Figure 5 about here]

With the synthetic control in hand, we contrast births observed in Recife relative to the synthetic control between April 2014-December 2017. We then implement a series of placebo tests (Abadie et al., 2014). These use the donor municipalities and substitute them for Recife, one at a time, to calculate the same quantities used in the main exercise issuing matched synthetic counterfactuals. These provide an empirical distribution of the effect estimates we would generate if "exposure" were randomly assigned to microregions in the working sample. We then present p-values for the estimates by using the distribution of computed effects normalized by the square-root of the mean prediction error. Conceptually, the placebo tests consider whether conclusions from the synthetic control tests are specific to Recife, or if we would have arrived at the same outcome by studying fertility in, for example, Belo Horizonte, a region known to have limited Zika exposure.

b) Difference-in-differences

The difference-in-differences extends the analysis to a broader contrast between regions with different levels of exposure to the epidemic for between January 2011 and December 2017. We use a regression model that tests for the differential evolution of birth cohorts month-by-month during and after the onset of the Zika epidemics. We first compare microregions within Pernambuco (including Recife) with others from across the country. We then expand the regions of interest to all of the microregions within the Northeastern Brazil. While very similar in principle to the analysis using synthetic control methods, this regression framework allows us to explore heterogeneity of impacts both within and across locations, enriching the ability to discuss the demographic and policy implications of our main findings.

The parsimonious difference-in-differences specification is:

$$ln(births)_{rym} = \gamma_r + \delta_{ym} + \theta_{rm} + \sum_{t=t_0}^{T} \alpha_{ym} \times treated_r + \sum_{t=t_0}^{T} \beta'_{ym} X_r + \epsilon_{rym} \quad (1)$$

where (log) births in microregion r in year-month ym, are explained by location fixed-effects (γ_r), time effects (δ_{ym}), microregion-specific seasonality (θ_{rm}, an interaction between a group indicator for the collection of microregions being scrutinized ($treated_r$) and time-effects covering the onset and aftermath of the epidemics (Jan 2015 onward). The parameter estimated for this interaction reflects the realization of a month-by-moth difference-in-differences estimator (measured as percentage change in birth cohorts due to the log transformation of the dependent variable). In an effort to reduce concerns regarding the comparability between exposed and comparison locations we also include interactions between time-effects between Jan-2015 and Dec-2017 with a microregion's economic characteristics (X_r), such as development level, inequality, and urbanization (all measured in 2010), as well as a red-flag indicator microregions with large infestation according to the LIRAa Index (measured in Jan-Feb and Oct-Nov of 2015).[21]

[21] We matched the municipality-level reports to the micro-regions in our data by taking the maximum value of the

Table A.3 (Supplementary materials) presents descriptive statistics for different groups of microregions analyzed in these specifications. It is clear from these that Pernambuco is an outlier in terms of microcephaly incidence (after Jun 2014) and high levels of mosquito infestation. The picture is less clear, but still troublesome for the whole Northeast region. Both are clearly disadvantaged along socio-economic dimensions relative to other areas of the country. Not surprisingly, underdevelopment is associated with higher exposure to the Zika epidemics and mosquito borne diseases more generally.

We study heterogenity across locations within the Northeast by including one additional interaction in term in the model. We do so to highlight the role of the mosquito infestation red-flagging as an important element in the salience of the message regarding disease risk. While the disease environment is obviously connected to the presence of the mosquito (conditional on the circulation of the virus), the information regarding higher concentration of *Aedes* larvae may have magnified the impact of the call to reassess fertility plans.

In a final piece of the analysis, we re-estimate Eq. 1 to predict variation in subgroups of the birth cohorts within locations. These include groups defined by maternal age at birth, educational attainment and access to private health providers. Attention to this variation helps to shed light on distinguishing impacts in the form of *quantum* or *tempo* effects, particularly if responses are observed among women closer to ending the most fecund portion of their life cycle. Socio-economic differences on the other hand can be informative of the mechanisms behind fertility fluctuations, in particular because poorer families are more likely exposed to diseases (and therefore its biological effects) but have less access to quality family planning services (and, therefore, less room for behavioral change). Because changes to fertility shift the composition of birth cohorts, these also shed light on the longer-run impacts of a short-run health shock on population patterns.

index across municipalities within a given microregion. The latter is red-flagged if at least one of the municipalities within it is red-flagged.

RESULTS

Fertility Change in Recife, Brazil

We begin by examining results of the synthetic control analysis focused on fertility in the center of the epidemic—Recife, Brazil. Effect estimates are presented for the entire period of study in Figure 6 and in Table 1. Infection risk in Recife is believed to have peaked in March 2015. If infection induced miscarriage among ongoing pregnancies, we would expect to see fertility rates fall between April and December 2015, depending on when in pregnancy excess miscarriage risk is highest. We see evidence of decline in birth rates relative to expected values in December 2015 but it does not extend to the month before and after (and is not statistically distinguishable from zero). The estimation indicates a a sizeable and significant reduction in birth rates in Recife from September 2016 (15.3%) to February 2017 (12.0%), indicating that reductions in fertility were sustained for approximately half a year, corresponding with pregnancy cohorts that would have been initiated between 1-5 months after the public health emergency was announced. The reduction peaks in November 2016. At that point, the birth cohorts were 19.9% smaller than predicted in the absence of the epidemic.

[Figure 6 and Table 1 about here]

We conducted several robustness checks. We re-estimated the synthetic control including Northeastern microregions outside Pernambuco. This increased the pool of “donor” microregions for comparison to 201 locations. We also tested alternative cut-off dates that expand the period of time during which the synthetic control is fit to March 2015 and November 2015 (versus March 2014). The latter is particularly different, since it matches Recife with a synthetic location inclusive of any potential early biological impact of Zika over fertility. We report the results in the supplementary online appendix; Table A.4 reports estimates and Table A.5 reports the composition of the synthetic control under the three alternatives. Results confirm October to December 2016 as the months during which gaps between observed birth rates and counterfactual levels are concentrated. Some of the effects are attenuated, suggesting that fertility variations beyond Recife and across the Northeast region are pertinent.

Fertility Change in Northeastern Brazil

To capture effects beyond Recife, we turn to the difference-in-differences estimates describing Pernambuco state and other locations across the Northeastern region of Brazil as "treated" (or exposed) microregions. We display the results in the form of month-to-month differential birth cohort sizes changes over time (in a log scale) between locations more and less exposed to Zika—as approximated by the infestation risk of the primary mosquito vector and cases of reported microcephally. We report estimates from 10 months before the public health emergency announcement to 25 months after it. Estimates describing the 18 microregions within Pernambuco are presented in Table 2 and Figure 7.

[Figure 7 and Table 2 about here]

The estimates indicate a small reduction in the birth rate in the immediate aftermath of the announcement, but most prominently between 9 to 17 months after it. These patterns are observed even after the interaction of additional controls for location characteristics interacted with time effects are included (Table 2, column [2]). They are also robust to the analysis which includes recorded miscarriages to the counts of births (Table 2, columns [3] and [4]), suggesting that fetal mortality accounted for in official records cannot explain variation in fertility we observe, particularly for the effects beyond the immediate aftermath of the announcement. The findings indicate that birth cohort sizes within Pernambuco were 25.8% smaller that in microregions outside the Northeast in October 2016, and that most of the cohorts born between 9 and 25 months after the public health emergency declaration are smaller in the former.

The use of difference-in-differences analysis also allows us to detect evidence of a differential behavioral response *within* the Pernambuco population. We re-estimate Eq. 1 for population subgroups: first by maternal education (less than high school versus high-school or more) and then by place of delivery (public versus private health facilities, which indexes SES in Brazil). The results are summarized in Figures 8 and 9 (Estimates provided in Table A.6 of the appendix). We find reductions for both higher- and lower-educated women, but the effects are substantively larger for women with more schooling and who give birth in private clinics. Eleven months after the public health emergency birth cohorts to educated women are 30.5% smaller than expected, while they are 19.6% smaller among less educated women, for example. The advantaged women/couples also seem to be responding earlier than their less-resourced counterparts, with significant reductions in births 9 months after national alerts about the Zika-microcephally link. Incidentally, this heterogeneity suggests that estimations of the effect of the epidemics using only data on deliveries computed by the public reimbursement system (SUS)—a potential resource for the scientific and policy communities because the SUS data allow observation of births closer to real time—will lead to underestimation. We confirmed this by examining patterns in deliveries in the SUS data (available from authors).

[Figure 8 about here]

[Figure 9 about here]

Turning to maternal age, we find clear evidence that the fertility response is smaller for women under 25 years of age, relative to older mothers, including those aged 35 or more. These are summarized in Figure 10 (estimates shown in online appendix Table A.7). The effects are 50-80% larger among older women. The effects are also more enduring for the oldest women (aged 35+); we see 10-20% reductions in birth cohort size relative to the expected value extend to November 2017, 24 months after the public health emergency was declared. Because this oldest group approaches ages of rapid declines in fecundity, the evidence is suggestive that Zika-related fertility delays may result in smaller completed fertility for a subset of the population.

[Figure 10 about here]

One implication of differences in effects across social groups is a change in the composition of birth cohorts with respect to parental characteristics. We examine changes in average composition of birth cohorts using the same difference-in-differences specifications used above. Table 3 summarizes the findings, and presents evidence of significant reductions in maternal education, access to private health care facilities and increases in age in the same months we observe an overall reduction in birth cohorts. We conclude that both size and compositional changes in birth cohorts are observed as result of behavioral changes triggered by the Zika epidemics in Pernambuco.

[Table 3 about here]

In a final piece of the analysis, we ask whether evidence of fertility changes extend into locations in the Northeast beyond the state of Pernambuco. We find quite clearly that the effects in terms of birth cohort sizes are much smaller than for this larger group of microregions (Figure 11 and Table 4). Moreover, when we examine heterogeneity across the region, mosquito infestation seem to play an important role in the fluctuations we observe in birth cohort sizes. Quite clearly, locations in which Zika transmitting mosquitos were more prevalent (as of 2015), and in which that information was made public by the red-flagging mechanism employed by the Brazilian Ministry of Health, have a stronger reduction in birth rates than elsewhere across the Northeast (Figure 12). We see very little evidence of difference between Panel A and B of Figure 12 during the months in which birth cohort size would have been a function of the biological effects of infection alone. As a result, the evidence in Figure 12 suggests that behavioral changes were linked to the perception of exposure to the new disease and, potentially, to the widespread perception of an inability to control the mosquito population by public officials.

[Figure 11 and Table 4 about here]

[Figure 12 about here]

DISCUSSION

Most theoretical models of human fertility predict the adjustment of reproductive behavior in response to changing circumstances (Hotz, Klerman, and Willis 1997, Johnson-Hanks et al., 2011, Trinitapoli and Yeatman 2018). Decades of demographic research document examples of changes in fertility rates in the wake of major events, including macroeconomic change, violence and conflict, environmental change, mortality shocks, and shifts in the disease landscape (Agadjanian and Prata 2002, Barreca, Deschenes, and Guldi 2018, Bleakley and Lange 2009, Heuveline and Poch 2007, Lindstrom Berhanu 2002, McCord, Conley, and Sachs 2017). In a few cases, evidence on accompanying changes to stated fertility intentions (e.g., Agadjanian and Prata 2002) suggests that some people may try to adjust fertility timing to align births with more favorable circumstances. In this study, we examine fertility change alongside the emergence of the Zika virus, a new, serious pregnancy health risk in the Americas. Although concerns have been raised about the potential biological effects of the epidemic, we find little evidence that the epidemic produced reduced the size of Brazilian birth cohorts via infection-induced reductions in fecundity. Instead we find evidence consistent with behavioral adjustments to delay fertility. Several characteristics of the epidemic support identification of these behavioral timing effects, in contrast to other competing explanations that often accompany large-scale threats to population health.

We examine fertility prior to and following the arrival of Zika in Brazil. By studying fertility effects of the epidemic in regions of the country where infection risk peaked before residents learned about it, we are able to distinguish biological and behavioral effects of disease spread. We find strong evidence of a behavioral response to *learning* about the disease. Attributable fertility declines are clustered 10-15 months after the national emergency was declared and the Brazilian Ministry of Health advised fertility delays.

We arrive at these conclusions by comparing the temporal patterns in fertility rates in the most directly affected regions against comparison regions that were affected later and to a lesser degree. The most affected regions include the epidemics epicenter, Recife, and other regions in Northeast Brazil. For the approaches used here—difference-in-differences and synthetic control analysis—

spillover effects of Zika exposure into comparison communities will bias estimates toward zero. During the period before information about Zika was widespread, we have no reason to believe spillover is an issue; models of the disease spread indicate that it was limited to Northeast Brazil in 2015 (e.g., Zhang 2017). For this reason, we expect that estimates of the biological effects of the disease are not downwardly biased by spillover effects. In the later periods, 2016 and 2017, the epidemic reached into other parts of Brazil. Risk and depictions of risk (reported *Aedes* mosquito infestation and reports on microcephaly) were still much higher in the Northeast. Nevertheless, it is possible that persons in Southern Brazil, for example, also made fertility timing adjustments, given the scale of media communication about the epidemic. If this is the case, the reductions of fertility we attribute here to learning about Zika may be even larger.

Pregnant women or couples intending to have children may have moved away from cities in the Northeast after learning about risks to infant health. Short-term migration would be a more drastic example of the kind of mosquito avoidance behaviors that have already been reported among women with higher socio-economic status in Recife (Marteleto et al. 2017). Since selective migration would complicate interpretation of our results, we used data on maternal place of residence in the analysis, instead of place of delivery; therefore, our estimates include short-term migrants as among the affected population. If departures are more permanent and include changes in residence, we might expect to see evidence of fertility reductions earlier in 2016, potentially as early as January or February 2016, as pregnant women exited Recife and other affected regions of Pernambuco and Northeastern Brazil. At that time, no information about trimester-specific risk was included in public health campaigns and women in the third trimester of pregnancy were also considered at risk. We do not see evidence of larger declines in births until late in 2016, long after migration would have started to be an option. When we investigate changes in birth cohorts based on birth location instead of location of maternal residence we see equivalent effects for the later portion of 2016 (columns [5] and [6] of Table 2). The effects in 2017 seem to be larger using this accounting of births than maternal residence, suggesting that some level of exodus from these regions happened among those with fertility plans. These differences, though, are not large enough to explain the majority of the effect estimates here.

In auxiliary exercises we also employed data from (a) school matriculation of 7 year-olds, and (b) electoral records, in order to provide evidence on the robustness of our results to migration concerns. They are not directly comparable to a sample of pregnant women but are the best estimate of short-term responses in form of migration. First, we look at families of school aged children and examine if children enrolled in Pernambuco schools in May 2015 (when the national census of schools is conducted by the Ministry of Education) are more likely to not be enrolled in a school within the state in May 2016, than children in other states outside the Northeast. We find no evidence of this. Second, making use of the fact that voting is compulsory in Brazil we examine election turnout as a measure of excess in absenteeism in Pernambuco microregions. We contrast elections in October 2010, 2012, 2014 and 2016 and find no evidence that the 2016 turnout rates in Pernambuco was atypical. These two results reinforce that the drop in pregnancy rates is unlikely a function of outmigration by women and couples considering pregnancy.

We find evidence of fertility timing effects among both more-resourced and less-resourced populations; the findings are substantively larger among older, better-educated women who use private health care. That fertility avoidance happened among less-resourced women is striking, given evidence that unintended pregnancy in Pernambuco is high and abortion is very difficult to access for low-income women.[22] Bahamondes and colleagues (2016) used data from pharmaceutical companies and found no evidence of increases in hormonal contraceptive sales in Brazil between 2014-2016. Some women may have avoided childbearing through other means, including barrier methods and delayed partnering. The findings here lend further support to demographic research documenting evidence of fertility adjustment outside of hormonal contraceptive methods (e.g., Goodkind 1993).

A number of scholars and policy analysts have highlighted Zika as a human rights issue for low-income women in high-risk areas (Rasanathan et al., 2017, Ribeiro et al., 2016). The important insights from these scholars underscore a more general observation about demographic behavior

[22]In auxiliary analysis using visits to SUS' health posts we find small increases in the number of Intra-Uterine Device (IUD) insertion procedures during February and March of 2016, nonetheless.

and population welfare. Temporal adjustment in fertility may (often) have distinct implications for population inequality. Indeed, we find evidence that the differences in fertility avoidance between less-educated and better-educated women was sufficient to shift the composition of birth cohorts in the later months of 2016 and beginning of 2017. Families who are least able to shift pregnancy timing are often those who must manage the heightened pregnancy risks with the fewest resources. While the net impact of temporal fertility adjustment on population welfare may be to improve the *average* birth conditions of the next generation, temporal adjustment will also exacerbate inequality—by increasing the association between birth conditions and parental resources.

In 2017, fertility rates in exposed regions returned to pre-epidemic levels, but not above them, indicating no evidence that fertility declines in 2016 and the beginning of 2017 were subsequently "recovered" or offset by the end of the period we can observe here. When we disaggregate fertility across age groups, we observe that declines are larger for older women. These delays may translate into forgone fertility as women approach the end of their reproductive ages. In coming years, studying completed fertility among older cohorts of women will help to clarify the longer-run effects of Zika on population change in Brazil.

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Tables and Figures for main text

Table 1: Differences in monthly crude birth rates (ln scale), synthetic control results.

Year month	Months since emergency declared	Difference Recife vs. Synthetic Recife	p-values	Events	Potential efffects reasoning
	[1]	[2]	[3]	[4]	[5]
2014m4	-19	-0.014	[0.782]		Biological (Fecundity/Miscarriage)
2014m5	-18	-0.018	[0.713]		Biological (Fecundity/Miscarriage)
2014m6	-17	-0.017	[0.745]		Biological (Fecundity/Miscarriage)
2014m7	-16	0.018	[0.660]		Biological (Fecundity/Miscarriage)
2014m8	-15	0.055	[0.303]		Biological (Fecundity/Miscarriage)
2014m9	-14	0.012	[0.814]		Biological (Fecundity/Miscarriage)
2014m10	-13	0.012	[0.888]		Biological (Fecundity/Miscarriage)
2014m11	-12	0.020	[0.798]		Biological (Fecundity/Miscarriage)
2014m12	-11	-0.019	[0.729]		Biological (Fecundity/Miscarriage)
2015m1	-10	-0.034	[0.436]		Biological (Fecundity/Miscarriage)
2015m2	-9	-0.051	[0.330]		Biological (Fecundity/Miscarriage)
2015m3	-8	-0.038	[0.473]	Peak of Zeak infection in NE	Biological (Fecundity/Miscarriage)
2015m4	-7	0.042	[0.447]		Biological (Fecundity/Miscarriage)
2015m5	-6	0.029	[0.537]		Biological (Fecundity/Miscarriage)
2015m6	-5	0.069	[0.165]		Biological (Fecundity/Miscarriage)
2015m7	-4	0.055	[0.293]		Biological (Fecundity/Miscarriage)
2015m8	-3	0.032	[0.585]		Biological (Fecundity/Miscarriage)
2015m9	-2	0.032	[0.548]		Biological (Fecundity/Miscarriage)
2015m10	-1	0.028	[0.585]		Biological (Fecundity/Miscarriage)
2015m11	0	-0.015	[0.830]	Health emergency declared	Biological (Fecundity/Miscarriage)
2015m12	1	-0.101	[0.128]		Biological (Fecundity/Miscarriage)
2016m1	2	-0.034	[0.543]		Biological (Fecundity/Miscarriage)
2016m2	3	-0.079	[0.122]		Biological (Fecundity/Miscarriage)
2016m3	4	-0.023	[0.676]		Biological + induced termination
2016m4	5	-0.019	[0.723]		Biological + induced termination
2016m5	6	0.046	[0.362]		Biological + induced termination
2016m6	7	0.005	[0.963]		Biological + induced termination
2016m7	8	0.015	[0.766]		Induced termination + pregnancy avoidance
2016m8	9	-0.036	[0.569]		Induced termination + pregnancy avoidance
2016m9	10	-0.153	[0.032]		Induced termination + pregnancy avoidance
2016m10	11	-0.183	[0.021]		Induced termination + pregnancy avoidance
2016m11	12	-0.198	[0.037]		Induced termination + pregnancy avoidance
2016m12	13	-0.144	[0.053]		Induced termination + pregnancy avoidance
2017m1	14	-0.109	[0.074]		Induced termination + pregnancy avoidance
2017m2	15	-0.120	[0.037]		Induced termination + pregnancy avoidance
2017m3	16	-0.076	[0.191]		Induced termination + pregnancy avoidance
2017m4	17	-0.027	[0.660]		Induced termination + pregnancy avoidance
2017m5	18	-0.010	[0.840]		Induced termination + pregnancy avoidance
2017m6	19	-0.028	[0.601]		Induced termination + pregnancy avoidance
2017m7	20	0.021	[0.729]		Induced termination + pregnancy avoidance
2017m8	21	0.067	[0.303]		Induced termination + pregnancy avoidance
2017m9	22	-0.046	[0.473]		Induced termination + pregnancy avoidance
2017m10	23	-0.017	[0.787]		Induced termination + pregnancy avoidance
2017m11	24	0.008	[0.899]		Induced termination + pregnancy avoidance
2017m12	25	-0.035	[0.564]		Induced termination + pregnancy avoidance

Notes: p-values are computed using exact randomization analogy. 187 microregions constitute the donor sample. Final synthetic control components listed in Table A.2.

Table 2: Difference in differences, Pernambuco versus Non-NE microregions.

months since	By location of maternal residence								By location of birth			
emergency	All live births				All births (including fetal deaths)				All live births		All births	
announced	[1]		[2]		[3]		[4]		[5]		[6]	
-10	-0.024	(0.023)	-0.051	(0.028)	-0.023	(0.023)	-0.048	(0.028)	-0.087	(0.051)	-0.083	(0.053)
-9	-0.008	(0.023)	0.012	(0.027)	-0.006	(0.024)	0.017	(0.028)	0.004	(0.045)	0.009	(0.045)
-8	-0.024	(0.026)	0.035	(0.030)	-0.024	(0.026)	0.034	(0.030)	-0.020	(0.074)	-0.019	(0.074)
-7	-0.022	(0.021)	-0.010	(0.028)	-0.020	(0.021)	-0.008	(0.028)	-0.030	(0.052)	-0.028	(0.052)
-6	-0.025	(0.020)	-0.013	(0.025)	-0.026	(0.021)	-0.014	(0.025)	-0.053	(0.084)	-0.056	(0.085)
-5	-0.026	(0.023)	0.008	(0.026)	-0.025	(0.023)	0.010	(0.026)	-0.015	(0.073)	-0.014	(0.073)
-4	0.014	(0.017)	0.030	(0.023)	0.016	(0.018)	0.033	(0.023)	0.026	(0.084)	0.030	(0.084)
-3	-0.031	(0.030)	-0.034	(0.031)	-0.028	(0.030)	-0.031	(0.031)	-0.077	(0.075)	-0.075	(0.075)
-2	-0.005	(0.025)	0.030	(0.028)	-0.006	(0.025)	0.029	(0.028)	0.000	(0.070)	-0.001	(0.070)
-1	-0.040	(0.029)	-0.038	(0.033)	-0.042	(0.028)	-0.040	(0.032)	-0.044	(0.092)	-0.046	(0.092)
0	-0.031	(0.022)	-0.026	(0.029)	-0.031	(0.022)	-0.027	(0.029)	-0.047	(0.086)	-0.048	(0.087)
1	-0.065	(0.029)	-0.063	(0.034)	-0.062	(0.029)	-0.061	(0.034)	-0.067	(0.079)	-0.065	(0.079)
2	-0.082	(0.024)	-0.071	(0.030)	-0.080	(0.025)	-0.067	(0.030)	-0.077	(0.070)	-0.072	(0.071)
3	-0.051	(0.022)	-0.037	(0.026)	-0.050	(0.023)	-0.035	(0.026)	-0.048	(0.065)	-0.045	(0.065)
4	-0.045	(0.023)	-0.006	(0.028)	-0.043	(0.023)	-0.003	(0.029)	0.042	(0.057)	0.046	(0.057)
5	-0.049	(0.029)	-0.052	(0.035)	-0.048	(0.029)	-0.051	(0.034)	-0.018	(0.063)	-0.018	(0.063)
6	0.010	(0.020)	0.045	(0.023)	0.009	(0.019)	0.045	(0.023)	0.097	(0.082)	0.097	(0.082)
7	-0.015	(0.025)	0.028	(0.026)	-0.018	(0.025)	0.026	(0.026)	0.078	(0.068)	0.076	(0.067)
8	0.004	(0.025)	0.022	(0.030)	0.000	(0.025)	0.019	(0.030)	0.073	(0.077)	0.071	(0.078)
9	-0.115	(0.026)	-0.101	(0.029)	-0.112	(0.025)	-0.098	(0.028)	-0.061	(0.081)	-0.059	(0.080)
10	-0.173	(0.027)	-0.175	(0.031)	-0.174	(0.027)	-0.176	(0.031)	-0.157	(0.072)	-0.158	(0.071)
11	-0.232	(0.028)	-0.258	(0.032)	-0.232	(0.027)	-0.258	(0.031)	-0.269	(0.109)	-0.272	(0.109)
12	-0.178	(0.025)	-0.176	(0.029)	-0.178	(0.025)	-0.173	(0.029)	-0.167	(0.094)	-0.165	(0.095)
13	-0.147	(0.020)	-0.159	(0.026)	-0.147	(0.020)	-0.159	(0.026)	-0.247	(0.088)	-0.248	(0.088)
14	-0.086	(0.027)	-0.129	(0.031)	-0.087	(0.027)	-0.131	(0.031)	-0.195	(0.089)	-0.193	(0.088)
15	-0.118	(0.026)	-0.100	(0.028)	-0.116	(0.027)	-0.100	(0.029)	-0.149	(0.068)	-0.149	(0.068)
16	-0.146	(0.029)	-0.122	(0.033)	-0.146	(0.028)	-0.120	(0.032)	-0.161	(0.081)	-0.158	(0.081)
17	-0.121	(0.019)	-0.106	(0.023)	-0.119	(0.018)	-0.105	(0.022)	-0.152	(0.067)	-0.152	(0.067)
18	-0.062	(0.016)	-0.040	(0.019)	-0.067	(0.016)	-0.044	(0.019)	-0.032	(0.065)	-0.033	(0.064)
19	-0.068	(0.028)	-0.037	(0.031)	-0.068	(0.029)	-0.038	(0.031)	-0.096	(0.097)	-0.097	(0.098)
20	-0.097	(0.033)	-0.095	(0.034)	-0.098	(0.032)	-0.096	(0.032)	-0.142	(0.118)	-0.144	(0.118)
21	-0.066	(0.031)	-0.046	(0.034)	-0.068	(0.031)	-0.046	(0.034)	-0.073	(0.094)	-0.072	(0.095)
22	-0.084	(0.033)	-0.094	(0.035)	-0.085	(0.034)	-0.094	(0.035)	-0.120	(0.085)	-0.120	(0.085)
23	-0.074	(0.028)	-0.074	(0.031)	-0.074	(0.029)	-0.073	(0.031)	-0.078	(0.079)	-0.079	(0.079)
24	-0.045	(0.033)	-0.046	(0.035)	-0.044	(0.032)	-0.045	(0.034)	-0.065	(0.084)	-0.065	(0.085)
25	-0.029	(0.026)	-0.046	(0.030)	-0.025	(0.027)	-0.044	(0.030)	-0.076	(0.089)	-0.075	(0.089)
Controls	NO		YES		NO		YES		YES		YES	

Notes: Standard-errors are clustered at the microregion level. There are 299 microregions and a total of 17,940 observations (covering Jan-2013 to Dec-2017). When included regression controls are post-Jan-15 time-effect interactions with local Human Development Index (HDI), urbanization rate, income inequality (Gini), mosquito infestation. All models include fixed-effects for microegion-specific seasonality, year-month and microregion.

Table 3: Difference in differences, composition of births, Pernambuco versus Non-NE microregions.

months since emergency announced	Share of live births to more educated moms [1]		Share of live births in private health facilities [2]		Average maternal age for live births [3]	
-10	0.007	(0.017)	0.011	(0.018)	0.071	(0.131)
-9	0.012	(0.016)	-0.005	(0.018)	-0.061	(0.137)
-8	-0.008	(0.013)	-0.007	(0.019)	-0.140	(0.091)
-7	-0.001	(0.011)	0.014	(0.017)	0.130	(0.174)
-6	-0.016	(0.013)	-0.002	(0.014)	0.076	(0.126)
-5	0.021	(0.014)	0.008	(0.019)	0.122	(0.123)
-4	0.011	(0.010)	0.005	(0.020)	-0.156	(0.141)
-3	-0.004	(0.012)	-0.010	(0.016)	0.086	(0.154)
-2	0.000	(0.016)	0.012	(0.020)	0.258	(0.119)
-1	-0.009	(0.014)	-0.020	(0.018)	0.062	(0.150)
0	-0.007	(0.013)	-0.016	(0.018)	-0.040	(0.184)
1	0.003	(0.015)	-0.021	(0.019)	-0.087	(0.135)
2	-0.014	(0.010)	-0.031	(0.023)	-0.242	(0.124)
3	0.016	(0.010)	-0.034	(0.029)	0.155	(0.110)
4	-0.020	(0.013)	-0.031	(0.024)	0.112	(0.112)
5	-0.008	(0.014)	-0.026	(0.021)	-0.099	(0.122)
6	-0.025	(0.014)	-0.042	(0.024)	0.046	(0.126)
7	-0.004	(0.012)	-0.031	(0.018)	0.093	(0.133)
8	0.013	(0.012)	-0.018	(0.023)	-0.176	(0.167)
9	-0.017	(0.014)	-0.048	(0.022)	-0.266	(0.161)
10	-0.027	(0.012)	-0.036	(0.021)	-0.393	(0.157)
11	-0.025	(0.010)	-0.052	(0.022)	-0.353	(0.141)
12	-0.013	(0.012)	-0.050	(0.024)	-0.541	(0.160)
13	-0.017	(0.014)	-0.059	(0.026)	-0.369	(0.130)
14	-0.026	(0.011)	-0.060	(0.031)	-0.317	(0.146)
15	-0.019	(0.015)	-0.065	(0.035)	-0.086	(0.139)
16	-0.035	(0.013)	-0.040	(0.026)	-0.254	(0.125)
17	-0.016	(0.012)	-0.052	(0.028)	-0.268	(0.153)
18	-0.020	(0.012)	-0.043	(0.032)	-0.191	(0.134)
19	0.003	(0.012)	-0.028	(0.024)	-0.074	(0.140)
20	0.019	(0.012)	-0.022	(0.023)	-0.215	(0.115)
21	-0.004	(0.014)	-0.004	(0.023)	-0.114	(0.144)
22	0.011	(0.019)	-0.006	(0.029)	0.180	(0.143)
23	-0.012	(0.016)	-0.006	(0.027)	-0.093	(0.127)
24	0.003	(0.014)	-0.011	(0.032)	-0.001	(0.156)
25	0.001	(0.013)	-0.014	(0.026)	0.005	(0.119)

Notes: Standard-errors are clustered at the microregion level. See notes in Table 2.

Table 4: Difference in differences, NE versus Non-NE microregions.

months since emergency announced	All NE [1]		Mosquito Infested NE [2]		Non Mosquito Infested NE [3]		Infested vs. non-Infested NE [4]		Infested vs. non-Infested NE [5]	
-10	-0.010	(0.013)	-0.022	(0.024)	-0.011	(0.014)	-0.011	(0.025)	-0.005	(0.026)
-9	0.012	(0.012)	0.019	(0.023)	0.009	(0.014)	0.010	(0.025)	0.012	(0.025)
-8	0.031	(0.011)	0.035	(0.021)	0.031	(0.012)	0.004	(0.023)	0.002	(0.024)
-7	0.004	(0.013)	-0.007	(0.029)	0.014	(0.014)	-0.021	(0.029)	-0.018	(0.029)
-6	-0.010	(0.011)	-0.031	(0.018)	-0.003	(0.012)	-0.027	(0.020)	-0.031	(0.020)
-5	0.013	(0.012)	0.012	(0.021)	0.014	(0.014)	-0.003	(0.023)	0.000	(0.023)
-4	0.025	(0.014)	0.035	(0.026)	0.019	(0.015)	0.016	(0.027)	0.012	(0.027)
-3	-0.007	(0.012)	-0.015	(0.021)	-0.006	(0.014)	-0.009	(0.023)	-0.007	(0.025)
-2	0.015	(0.011)	0.010	(0.025)	0.016	(0.012)	-0.006	(0.026)	-0.009	(0.027)
-1	0.016	(0.012)	-0.007	(0.025)	0.024	(0.014)	-0.031	(0.026)	-0.029	(0.028)
0	0.007	(0.015)	-0.017	(0.033)	0.015	(0.016)	-0.031	(0.034)	-0.031	(0.035)
1	-0.022	(0.013)	-0.021	(0.024)	-0.019	(0.015)	-0.002	(0.026)	0.004	(0.028)
2	-0.004	(0.013)	-0.025	(0.029)	0.000	(0.014)	-0.025	(0.029)	-0.020	(0.030)
3	-0.020	(0.013)	-0.014	(0.027)	-0.019	(0.014)	0.005	(0.028)	0.009	(0.028)
4	0.016	(0.013)	0.008	(0.026)	0.019	(0.014)	-0.011	(0.027)	-0.004	(0.028)
5	0.005	(0.012)	-0.014	(0.025)	0.009	(0.014)	-0.024	(0.028)	-0.010	(0.028)
6	0.019	(0.015)	0.018	(0.022)	0.017	(0.016)	0.001	(0.024)	-0.002	(0.024)
7	0.029	(0.013)	0.040	(0.020)	0.026	(0.015)	0.014	(0.023)	0.016	(0.024)
8	0.045	(0.014)	0.033	(0.025)	0.048	(0.016)	-0.015	(0.027)	-0.018	(0.028)
9	-0.036	(0.014)	-0.055	(0.024)	-0.025	(0.015)	-0.030	(0.026)	-0.029	(0.027)
10	-0.068	(0.014)	-0.159	(0.026)	-0.033	(0.014)	-0.125	(0.028)	-0.124	(0.028)
11	-0.080	(0.014)	-0.133	(0.026)	-0.060	(0.016)	-0.073	(0.029)	-0.062	(0.029)
12	-0.053	(0.014)	-0.112	(0.029)	-0.027	(0.016)	-0.085	(0.031)	-0.078	(0.032)
13	-0.052	(0.015)	-0.108	(0.026)	-0.025	(0.017)	-0.082	(0.029)	-0.081	(0.030)
14	-0.041	(0.015)	-0.096	(0.027)	-0.030	(0.017)	-0.066	(0.030)	-0.057	(0.031)
15	-0.030	(0.015)	-0.060	(0.024)	-0.021	(0.017)	-0.040	(0.026)	-0.038	(0.026)
16	-0.062	(0.012)	-0.093	(0.023)	-0.050	(0.014)	-0.043	(0.026)	-0.041	(0.027)
17	-0.015	(0.013)	-0.047	(0.025)	0.000	(0.014)	-0.047	(0.027)	-0.041	(0.027)
18	-0.010	(0.012)	-0.019	(0.019)	-0.009	(0.013)	-0.010	(0.021)	-0.014	(0.021)
19	-0.008	(0.013)	-0.001	(0.024)	-0.012	(0.015)	0.011	(0.027)	0.015	(0.027)
20	-0.035	(0.013)	-0.038	(0.025)	-0.034	(0.015)	-0.004	(0.026)	-0.003	(0.028)
21	-0.013	(0.013)	-0.002	(0.023)	-0.016	(0.015)	0.014	(0.026)	0.014	(0.028)
22	-0.041	(0.015)	-0.053	(0.023)	-0.042	(0.017)	-0.011	(0.026)	-0.013	(0.027)
23	-0.004	(0.014)	0.005	(0.026)	-0.007	(0.015)	0.012	(0.026)	0.012	(0.027)
24	0.019	(0.015)	0.014	(0.023)	0.016	(0.017)	-0.002	(0.026)	0.007	(0.027)
25	0.002	(0.015)	0.019	(0.021)	-0.001	(0.018)	0.020	(0.025)	0.023	(0.026)

Notes: Standard-errors are clustered at the microregion level. See notes in Table 2.

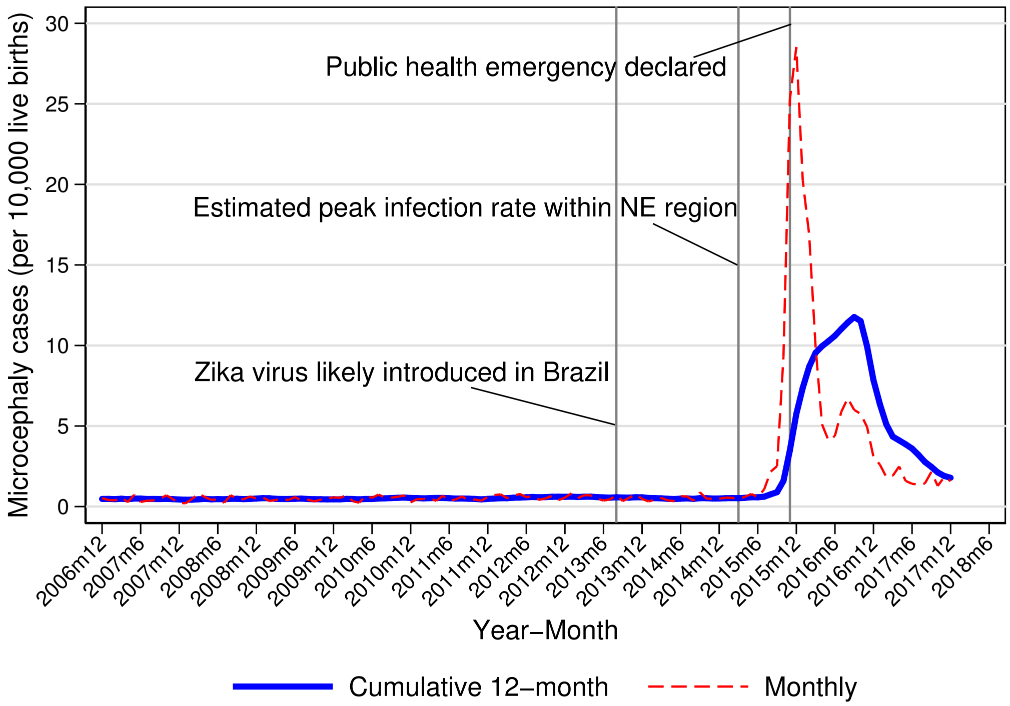

Figure 1: Reported microcephally cases per 10,000 live births.
[Source: SINASC-DATASUS]

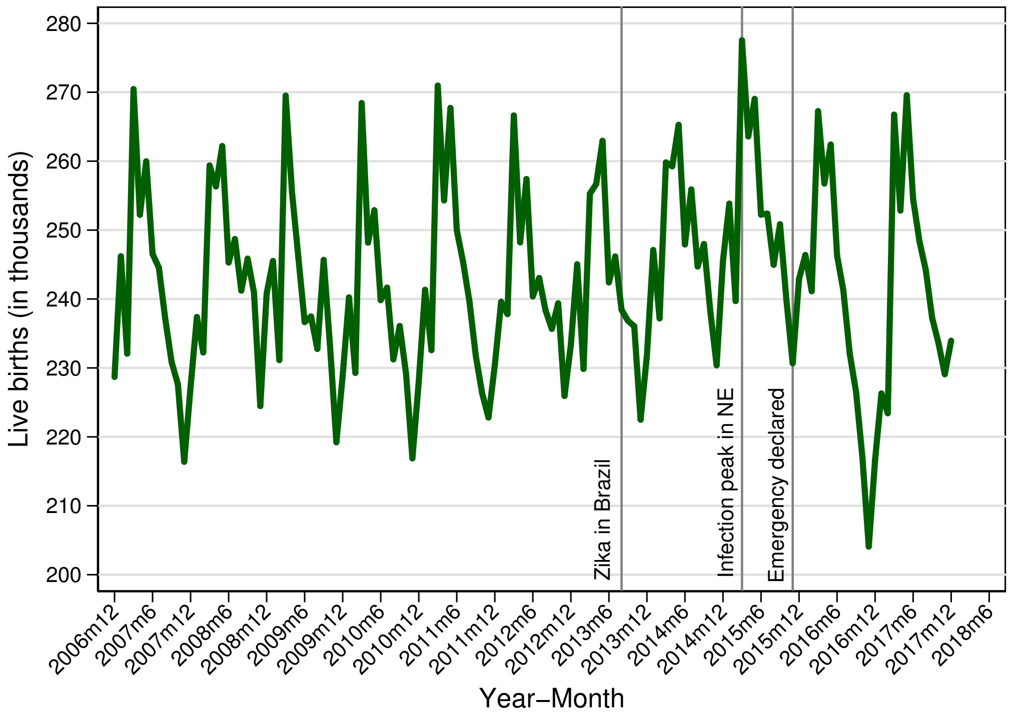

Figure 2: Live births per month - Brazil

[Source: SINASC-DATASUS]

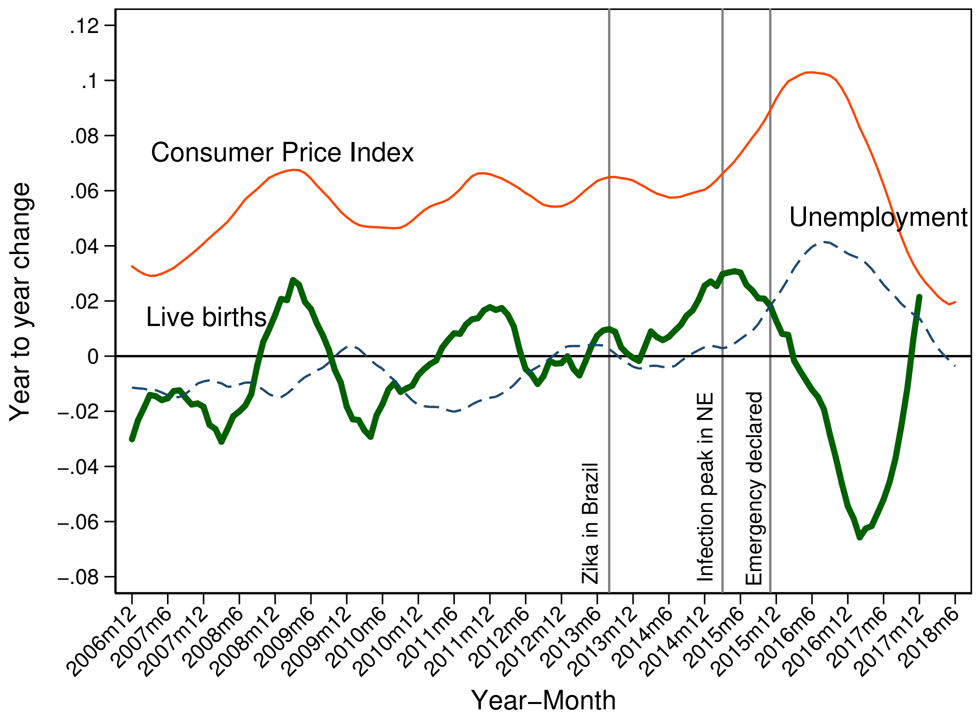

Figure 3: Year-to-year change in 12-month birth cohorts, consumer price index, and unemployment rate

[Source: SINASC-DATASUS and IPEADATA]

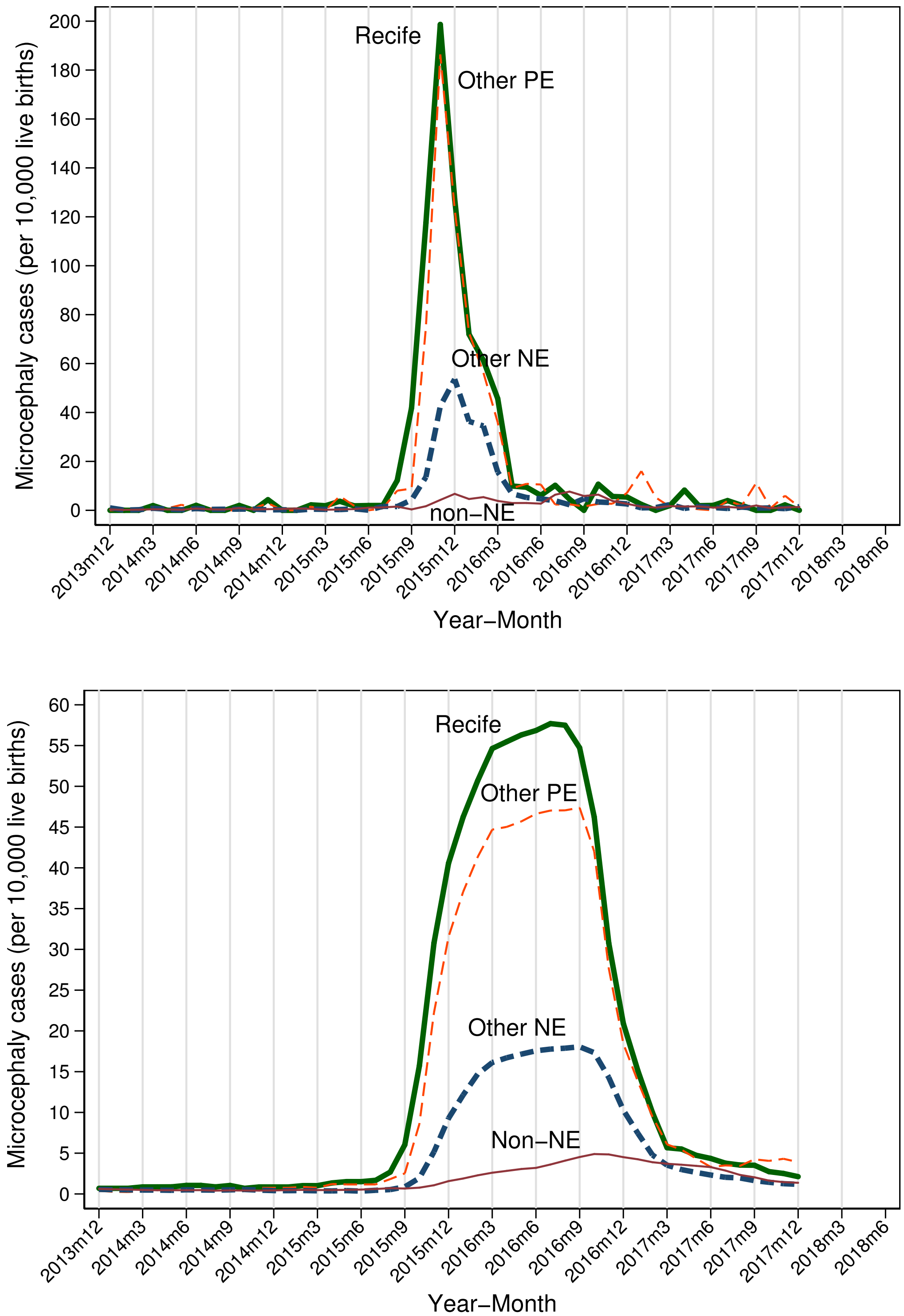

Figure 4: Reported microcephally cases per 10,000 live births, Monthly (top) and 12-month Aggregate (bottom) rates.

For the spatial distribution of these microregions see Figures A.1 and A.2 on the Appendix. [Source: SINASC-DATASUS]

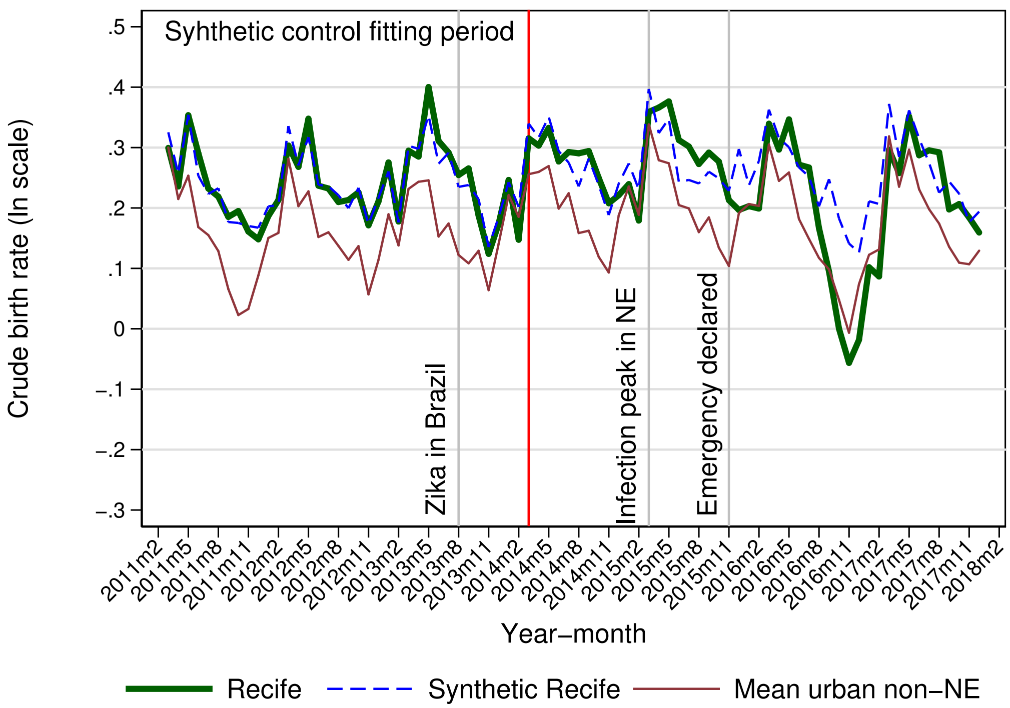

Figure 5: Monthly Crude Birth Rate (Relative to 2010 population, ln scale).

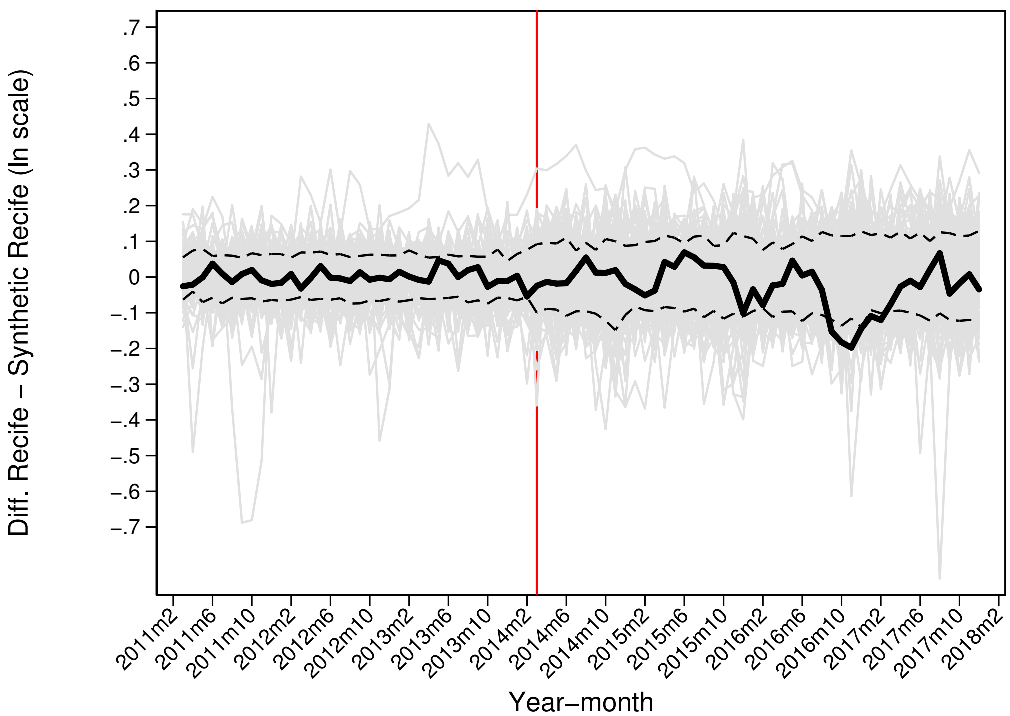

Figure 6: Crude Live Births Rate (births per 1,000 inhabitants in 2010): Recife vs. Synthetic Recife, ln scale.

Differences in gray produced from 187 placebo estimations. Dashed lines represent the top and bottom 10% of the placebo-generated differences. Vertical line indicates last observation in fitting algorithm. Projections are to the right of this marker.

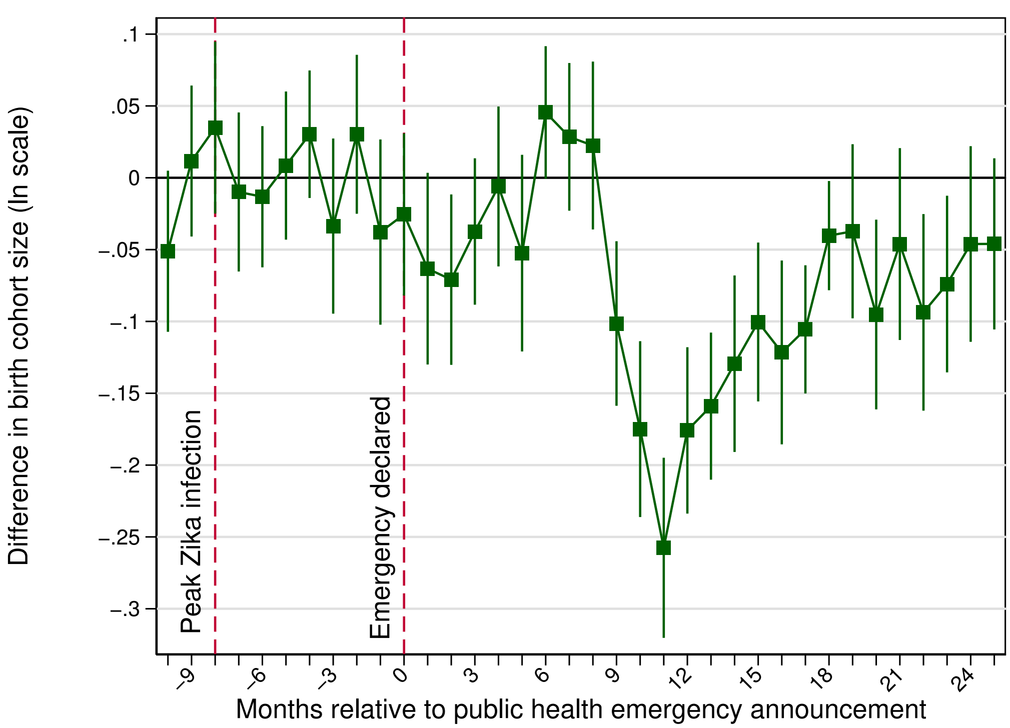

Figure 7: Difference in differences, live birth cohorts, Pernambuco versus Non-NE microregions.

Notes: Reported 95% confidence intervals are based on standard-errors clustered at the micorregion level. Regression controls for time interactions with local Human Development Index (HDI), urbanization rate, income inequality (Gini), mosquito infestation, and fixed-effects for microegion-specific seasonality, year-month and microregion.

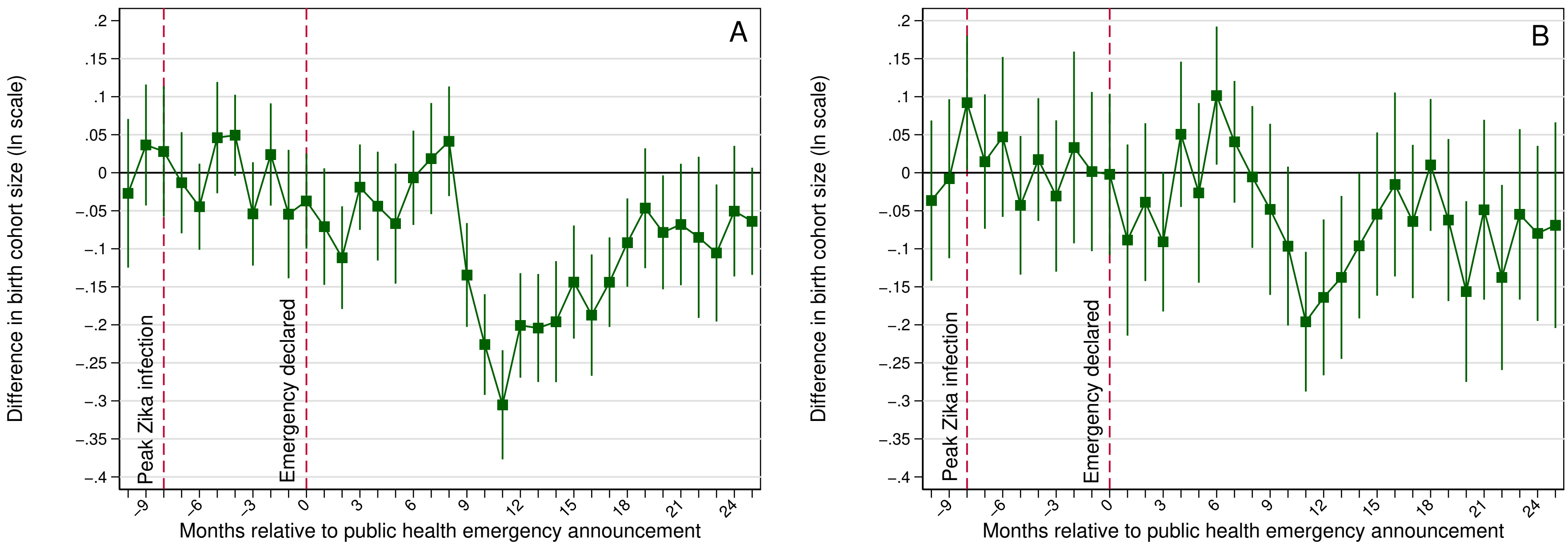

Figure 8: Difference in differences, live birth cohorts, Pernambuco versus Non-NE microregions, by maternal level of education: Completed Primary or more (Panel A) and Incomplete Primary or less (Panel B)

Notes: See notes in Figure 7.

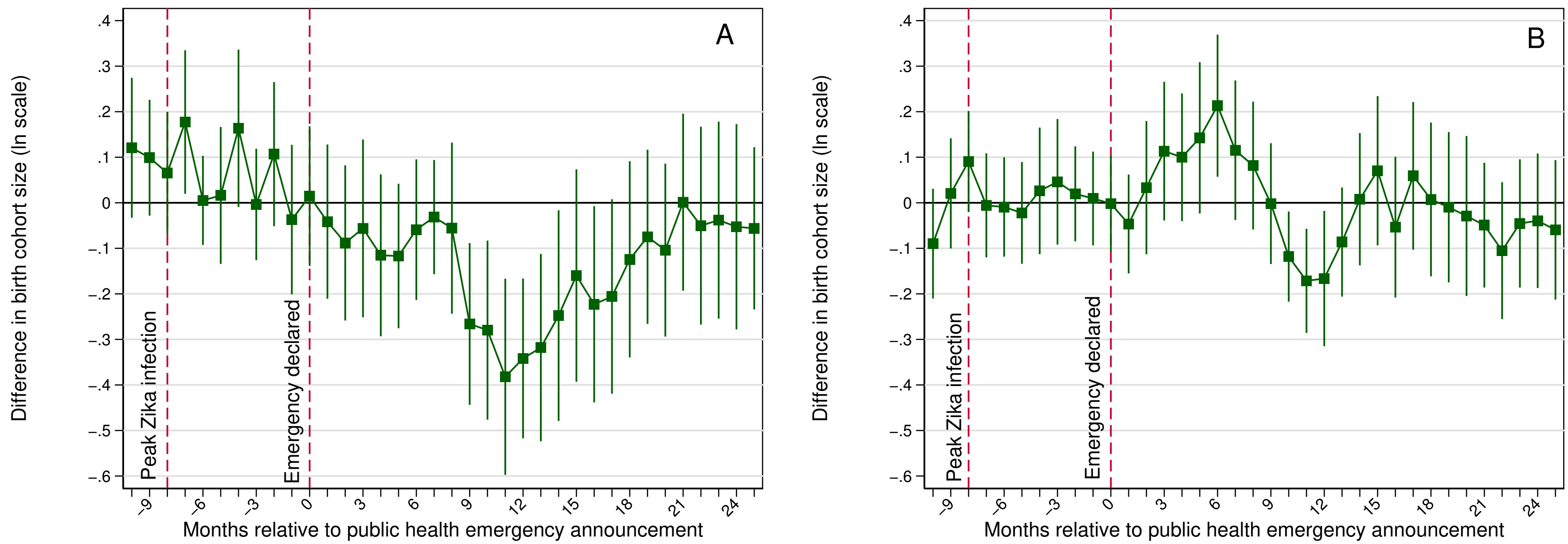

Figure 9: Difference in differences, live birth cohorts, Pernambuco versus Non-NE microregions, by type of health care utilized at delivery: Private (Panel A) and Public (Panel B)

Notes: See notes in Figure 7.

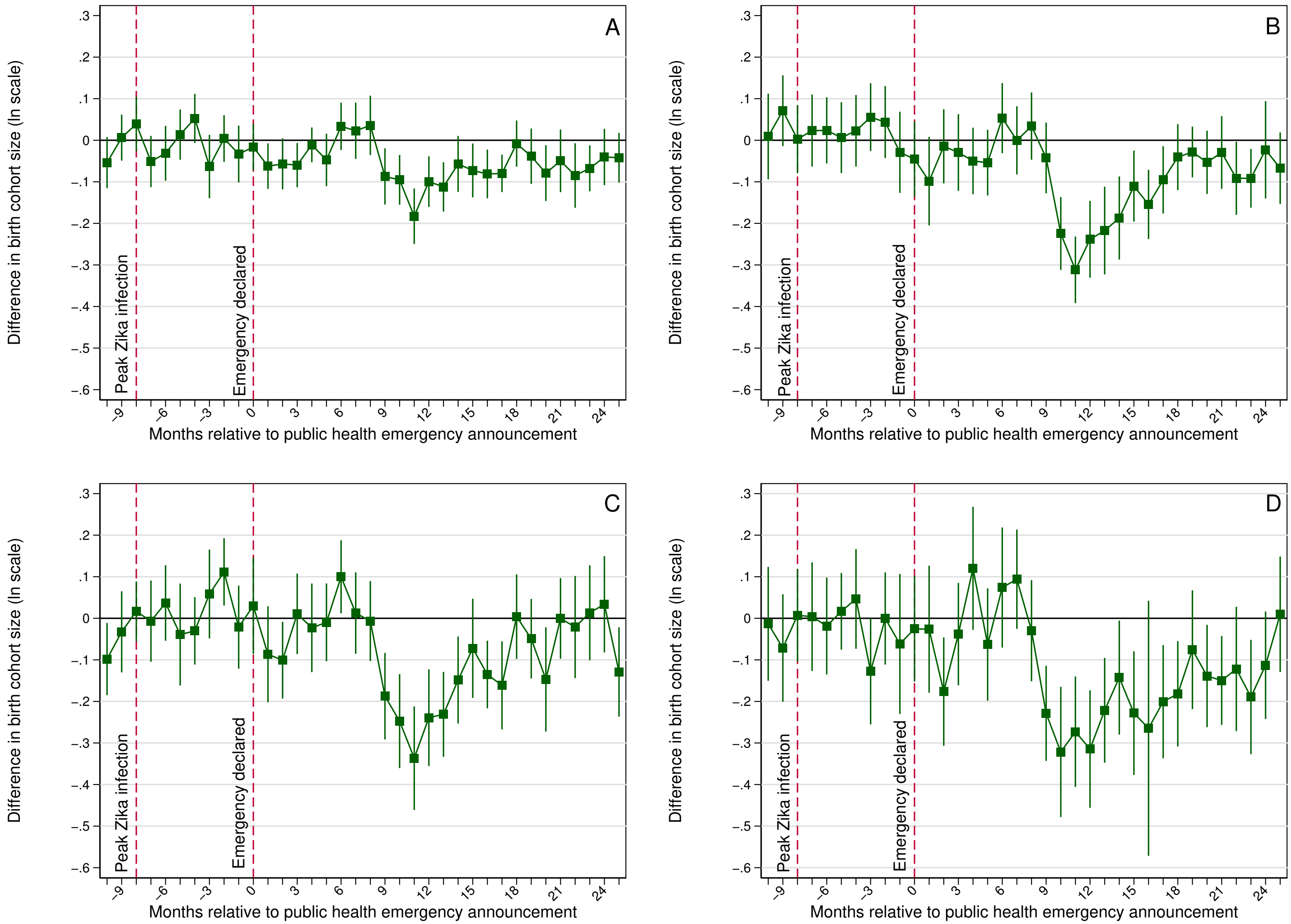

Figure 10: Difference in differences, live birth cohorts, Pernambuco versus Non-NE microregions, by maternal age group: Under 25 (Panel A), 25 to 29 (Panel B), 30 to 34 (Panel C) and 35 and older (Panel D)

Notes: See notes in Figure 7.

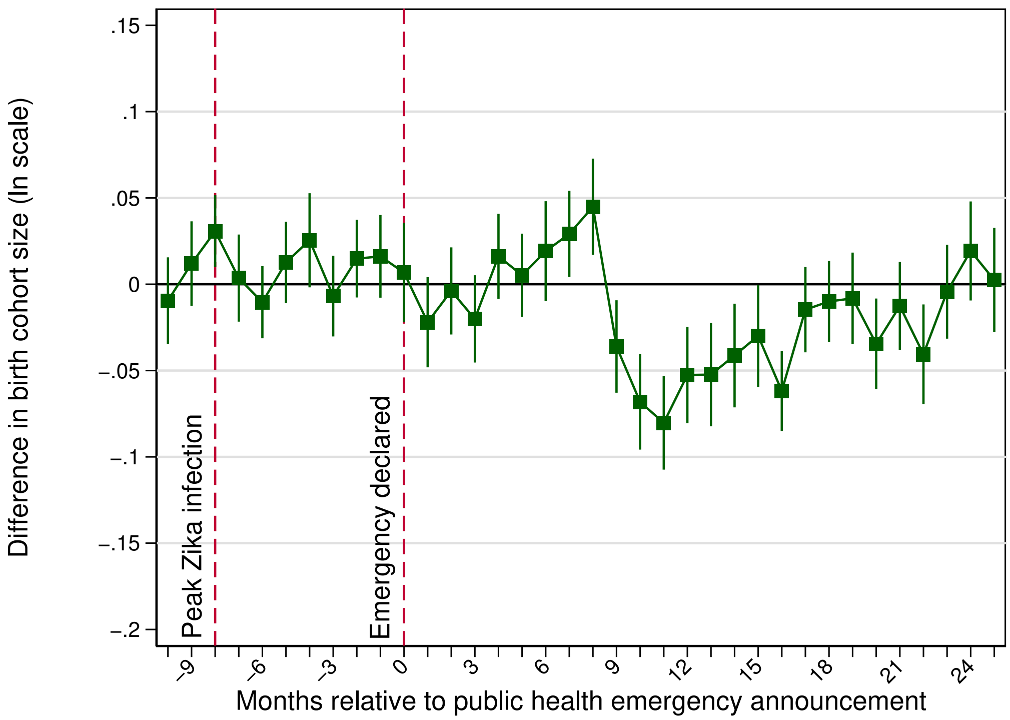

Figure 11: Difference in differences, live birth cohorts, NE versus Non-NE microregions.

Notes: See notes in Figure 7.

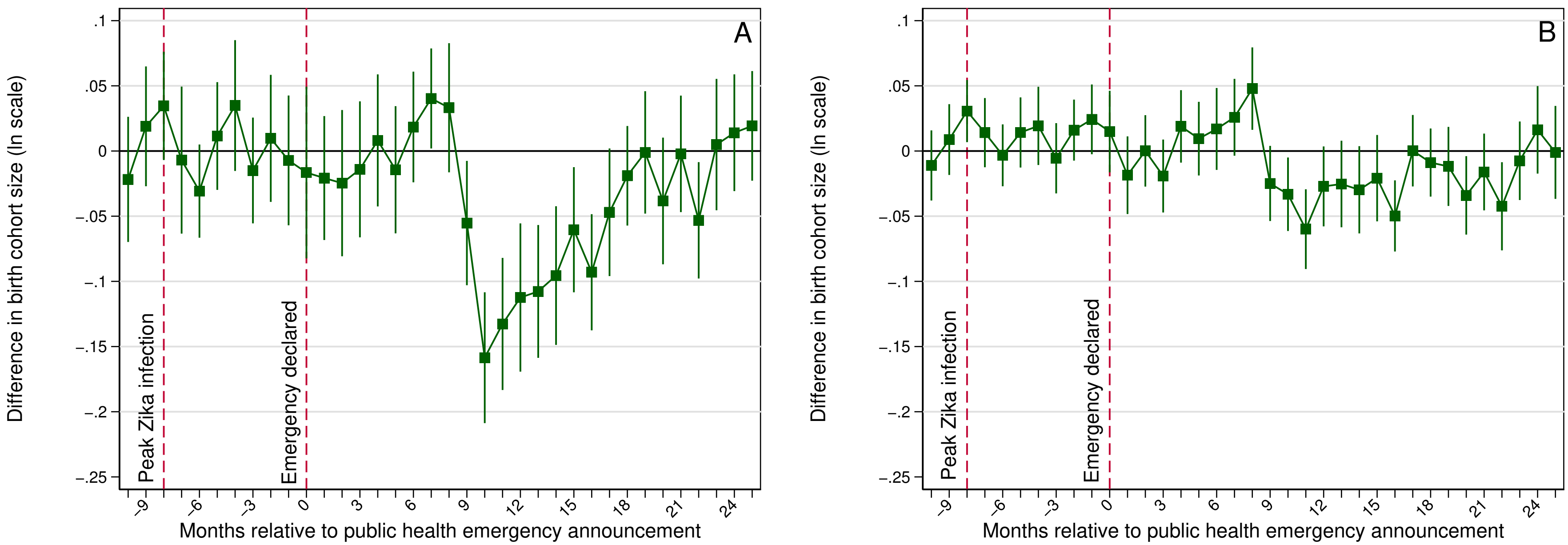

Figure 12: Difference in differences, live birth cohorts, NE versus Non-NE microregions, by type LIRAa Index red-flag: High mosquito infestation (Panel A) and Low/unknown mosquito infestation (Panel B)

Notes: See notes in Figure 7.

Tables and Figures – Appendix

Table A.1: **Descriptive statistics for covariates in synthetic control analysis.**

	Recife	Microregions outside NE		Synthetic Recife
	[1]	[2]		[3]
Socio-economic indicators				
Human Development Index (HDI)	0.730	0.739	(0.0021)	0.730
HDI - income	0.720	0.732	(0.0024)	0.731
HDI - health	0.820	0.842	(0.0012)	0.839
HDI - education	0.661	0.656	(0.0033)	0.637
Gini coefficient	0.608	0.507	(0.0029)	0.512
Share population in urban areas	0.969	0.879	(0.0043)	0.863
Demographic composition				
Population share of females born 1971-1975	0.043	0.037	(0.0002)	0.037
Population share of females born 1976-1980	0.046	0.041	(0.0002)	0.041
Population share of females born 1981-1985	0.048	0.043	(0.0003)	0.044
Population share of females born 1986-1990	0.046	0.043	(0.0003)	0.045
Population share of females born 1991-1995	0.042	0.043	(0.0003)	0.045
Population share of females born 1996-2000	0.041	0.042	(0.0003)	0.044
Population share of females born after 2000	0.071	0.069	(0.0007)	0.073
Socio-demographic indicators for births				
Share of births to moms 35 and older	0.096	0.105	(0.0017)	0.092
Share of births to moms 30 to 34	0.176	0.181	(0.0018)	0.169
Share of births to moms 25 to 29	0.256	0.258	(0.0011)	0.255
Share of births to moms under 24	0.472	0.456	(0.0038)	0.485
Average maternal age	25.558	25.828	(0.0578)	25.394
Share of births who are boys	0.512	0.512	(0.0007)	0.509
Share of births who are black of brown	0.724	0.295	(0.0182)	0.423
Share of births who are white	0.275	0.670	(0.0199)	0.548
Share of births to moms with high-school or more	0.180	0.196	(0.0036)	0.162
Share of births to moms with 8 to 11 years of ed.	0.499	0.500	(0.0054)	0.512
Share of births to moms with 4 to 7 years of ed.	0.267	0.247	(0.0041)	0.246
Share of births to moms with 3 years of ed. or less	0.048	0.044	(0.0017)	0.052
Share of births to moms with unknown ed. level	0.005	0.013	(0.0015)	0.029
Share of births in private health units	0.618	0.771	(0.0172)	0.709
Birth outcomes				
Microcephaly cases per 10,000	0.560	0.580	(0.1000)	0.320
Congenital anomaly cases per 10,000	93.930	73.620	(2.0400)	68.300
Share of births before term	0.076	0.075	(0.0012)	0.067
Share with very low birth weight	0.015	0.013	(0.0002)	0.011
Share with low birth weight	0.088	0.089	(0.0008)	0.080
Microregions	1	187		1

Notes: Standard-errors computed using jacknife for sample of microregions outside the Northeast (NE). All indicators reflect measures for year 2010.

Table A.2: Microregional cmposition of Synthetic Recife.

Microregion (Regiao Geografica Imediata)	State	Unit weight
Amparo	SP	0.102
Anapolis	GO	0.009
Andradina	SP	0.061
Araguaina	TO	0.044
Belem	PA	0.005
Brusque	SC	0.110
Charqueadas - Triunfo - Sao Jeronimo	RS	0.038
Cianorte	PR	0.010
Criciuma	SC	0.018
Dores do Indaia	MG	0.091
Inhumas - Itabera - Anicuns	GO	0.029
Ituiutaba	MG	0.089
Joao Monlevade	MG	0.012
Palmeiras de Goias	GO	0.007
Parauapebas	PA	0.031
Porangatu	GO	0.014
Porto Velho	RO	0.201
Rio Branco	AC	0.093
Sao Lourenco	MG	0.029
Sao Sebastiao do Paraiso	MG	0.008

Note: Synthetic-Recife from pool of 187 donor microregions and fitting period Jan 2011-Mar 2014.

Table A.3: Table descriptive statistics for difference-in-differences analysis.

	Pernambuco State [1]	Northeast region (including PE) [2]	Non-Northeast [3]	Pernambuco vs non-NE [4]	Northeast vs non-NE [5]
Composition					
Number of microregions	18	131	281	299	412
Average number of municipalities within a microregion	10.28	12.43	11.58	-1.31	0.84
	(1.364)	(0.615)	(0.408)	(1.424)	(0.739)
Disease environment					
Average peak monthly microcephaly rate (before Jun-14)	28.50	20.41	22.89	5.61	-2.48
	(4.132)	(1.865)	(1.545)	(4.412)	(2.422)
Average peak monthly microcephaly rate (after Jun-14)	218.356	116.283	39.642	178.715	76.641
	(23.689)	(7.497)	(2.518)	(23.823)	(7.908)
Share of microregions with high LIRAa	1.00	0.71	0.64	0.36	0.07
	-	(0.040)	(0.029)	(0.029)	(0.049)
Development stage – economic conditions					
Average HDI	0.62	0.61	0.71	-0.09	-0.10
	(0.009)	(0.004)	(0.003)	(0.010)	(0.005)
Average HDI-income	0.60	0.59	0.70	-0.11	-0.12
	(0.010)	(0.004)	(0.004)	(0.010)	(0.005)
Average HDI-health	0.77	0.76	0.83	-0.06	-0.07
	(0.005)	(0.002)	(0.002)	(0.005)	(0.002)
Average HDI-education	0.52	0.52	0.62	-0.10	-0.10
	(0.012)	(0.005)	(0.005)	(0.013)	(0.007)
Average Gini	0.54	0.55	0.52	0.02	0.03
	(0.009)	(0.003)	(0.003)	(0.009)	(0.004)
Average urbanization rate	0.68	0.63	0.80	-0.12	-0.17
	(0.030)	(0.011)	(0.008)	(0.031)	(0.014)
Demographic composition					
Average mom-age	24.76	24.54	25.47	-0.71	-0.93
	(0.129)	(0.071)	(0.065)	(0.144)	(0.097)
Average share of moms with HS or more	0.12	0.11	0.18	-0.06	-0.07
	(0.006)	(0.004)	(0.004)	(0.007)	(0.005)
Average share of moms 8 to 11 years of schooling	0.37	0.37	0.47	-0.10	-0.09
	(0.014)	(0.006)	(0.005)	(0.015)	(0.008)

Notes: Standard-errors are based on jacknife method. Microcephally rates are per 10,000 births. all indicators (except disease environment) measured in 2010.

Table A.4: Robustness exercises – Differences in monthly crude birth rates (ln scale), synthetic control results.

Year month	Months since emergency declared	Alternative donor sample [1]	p-values [2]	Cutoff in Mar-2015 [3]	p-values [4]	Cutoff in Nov-2015 [5]	p-values [6]
2014m4	-19	-0.008	[0.866]	-0.024	[0.548]	-0.031	[0.489]
2014m5	-18	-0.014	[0.757]	-0.003	[0.936]	-0.007	[0.814]
2014m6	-17	-0.014	[0.723]	0.013	[0.739]	0.000	[1.000]
2014m7	-16	-0.023	[0.535]	-0.010	[0.734]	-0.026	[0.500]
2014m8	-15	0.010	[0.847]	0.040	[0.287]	0.031	[0.426]
2014m9	-14	0.005	[0.906]	0.034	[0.378]	0.029	[0.431]
2014m10	-13	-0.003	[0.955]	0.013	[0.814]	0.005	[0.915]
2014m11	-12	-0.011	[0.851]	0.014	[0.777]	0.009	[0.888]
2014m12	-11	-0.020	[0.673]	-0.032	[0.436]	-0.039	[0.356]
2015m1	-10	-0.054	[0.193]	-0.054	[0.191]	-0.058	[0.154]
2015m2	-9	-0.050	[0.302]	-0.024	[0.500]	-0.034	[0.362]
2015m3	-8	-0.029	[0.510]	-0.004	[0.915]	-0.019	[0.660]
2015m4	-7	0.021	[0.644]	0.044	[0.324]	0.028	[0.473]
2015m5	-6	0.020	[0.663]	0.054	[0.223]	0.036	[0.346]
2015m6	-5	0.015	[0.728]	0.039	[0.410]	0.027	[0.505]
2015m7	-4	-0.008	[0.851]	0.059	[0.223]	0.043	[0.266]
2015m8	-3	0.017	[0.743]	0.012	[0.809]	0.005	[0.856]
2015m9	-2	-0.008	[0.876]	0.014	[0.809]	0.006	[0.910]
2015m10	-1	0.014	[0.743]	0.018	[0.713]	0.012	[0.750]
2015m11	0	-0.014	[0.782]	0.009	[0.910]	0.000	[0.973]
2015m12	1	-0.053	[0.297]	-0.057	[0.293]	-0.062	[0.239]
2016m1	2	-0.023	[0.634]	-0.061	[0.218]	-0.059	[0.218]
2016m2	3	-0.033	[0.416]	-0.061	[0.165]	-0.069	[0.144]
2016m3	4	-0.005	[0.901]	0.012	[0.803]	-0.001	[0.989]
2016m4	5	-0.005	[0.926]	0.000	[1.000]	-0.004	[0.920]
2016m5	6	-0.005	[0.931]	0.064	[0.191]	0.050	[0.309]
2016m6	7	-0.024	[0.599]	0.053	[0.314]	0.044	[0.404]
2016m7	8	-0.023	[0.634]	0.064	[0.298]	0.051	[0.383]
2016m8	9	-0.062	[0.243]	-0.005	[0.899]	-0.019	[0.649]
2016m9	10	-0.123	[0.045]	-0.074	[0.218]	-0.082	[0.160]
2016m10	11	-0.165	[0.025]	-0.128	[0.048]	-0.139	[0.027]
2016m11	12	-0.150	[0.040]	-0.125	[0.085]	-0.143	[0.048]
2016m12	13	-0.159	[0.015]	-0.158	[0.011]	-0.164	[0.011]
2017m1	14	-0.071	[0.188]	-0.060	[0.293]	-0.068	[0.250]
2017m2	15	-0.061	[0.218]	-0.051	[0.346]	-0.063	[0.229]
2017m3	16	-0.037	[0.460]	-0.010	[0.856]	-0.029	[0.606]
2017m4	17	-0.020	[0.683]	-0.011	[0.835]	-0.026	[0.580]
2017m5	18	0.000	[0.995]	0.012	[0.824]	0.000	[0.995]
2017m6	19	-0.023	[0.599]	0.018	[0.793]	0.007	[0.915]
2017m7	20	0.003	[0.931]	0.061	[0.340]	0.052	[0.388]
2017m8	21	0.032	[0.545]	0.053	[0.383]	0.050	[0.404]
2017m9	22	-0.043	[0.436]	-0.004	[0.941]	-0.015	[0.761]
2017m10	23	-0.012	[0.812]	0.011	[0.851]	0.004	[0.941]
2017m11	24	-0.009	[0.851]	0.004	[0.968]	0.006	[0.926]
2017m12	25	-0.040	[0.455]	-0.020	[0.777]	-0.031	[0.596]

Notes: p-values are computed using exact randomization analogy. 201 microregions constitute the donor sample in columns [1] and [2]. Final synthetic control components listed in Table A.5.

Table A.5: Microregions that compose alternatives for construction of Synthetic Recife.

Microregion	weight	Microregion	weight	Microregion	weight
(Regiao Geografica Imediata)					
from pool of 201		match Mar-2105		match Nov-2105	
Amparo	0.039	Adamantina - Lucelia	0.019	Araguaina	0.006
Anapolis	0.031	Belem	0.261	Belem	0.196
Andradina	0.047	Belo Horizonte	0.092	Belo Horizonte	0.104
Assis	0.030	Caldas Novas-Morrinhos	0.017	Caldas Novas-Morrinhos	0.009
Avare	0.007	Campos dos Goytacazes	0.030	Campos dos Goytacazes	0.041
Brusque	0.019	Cianorte	0.001	Cianorte	0.014
Ceres - Rialma Goianesia	0.021	Dores do Indaia	0.008	Franca	0.024
Charqueadas - Triunfo - Sao Jeronimo	0.032	Formiga	0.002	Gurupi	0.072
Criciuma	0.011	Gurupi	0.038	Joao Monlevade	0.077
Dores do Indai	0.027	Joao Monlevade	0.065	Marilia	0.027
Fortaleza	0.200	Juiz de Fora	0.003	Montes Claros	0.010
Maceio	0.099	Marilia	0.043	Palmas	0.094
Natal	0.045	Montes Claros	0.018	Palmeiras de Goias	0.027
Palmeiras de Goias	0.042	Palmas	0.091	Rio Bonito	0.021
Paranagua	0.010	Rio Bonito	0.032	Rio Branco	0.098
Paranaiba - Chapadao do Sul - Cassilandia	0.008	Rio Branco	0.093	Rondonopolis	0.015
Parauapebas	0.103	Rondonopolis	0.015	Santo Antonio de Padua	0.008
Pocos de Caldas	0.018	Santo Antonio de Padua	0.018	Volta Redonda - Barra Mansa	0.160
Porto Velho	0.006	Volta Redonda - Barra Mansa	0.155		
Salvador	0.096				
Sao Lourenco	0.070				
Sao Luis	0.041				

Table A.6: Difference in differences, by maternal education and utilization of private health care facilities, Pernambuco versus Non-NE microregions.

months since emergency announced	Live births to more educated mothers [1]		Live births to less educated mothers [2]		Live births in private facilities [3]		Live births in non-private facilities [4]	
-10	-0.027	(0.050)	-0.037	(0.054)	0.121	(0.078)	-0.090	(0.061)
-9	0.036	(0.040)	-0.008	(0.053)	0.099	(0.065)	0.021	(0.061)
-8	0.028	(0.044)	0.092	(0.045)	0.066	(0.068)	0.091	(0.056)
-7	-0.013	(0.034)	0.015	(0.045)	0.177	(0.080)	-0.006	(0.058)
-6	-0.045	(0.029)	0.047	(0.053)	0.005	(0.050)	-0.009	(0.055)
-5	0.046	(0.037)	-0.043	(0.046)	0.016	(0.076)	-0.022	(0.057)
-4	0.049	(0.027)	0.017	(0.041)	0.164	(0.088)	0.026	(0.071)
-3	-0.054	(0.035)	-0.031	(0.051)	-0.004	(0.062)	0.046	(0.070)
-2	0.024	(0.034)	0.033	(0.064)	0.107	(0.080)	0.020	(0.053)
-1	-0.054	(0.043)	0.002	(0.053)	-0.037	(0.083)	0.010	(0.052)
0	-0.037	(0.032)	-0.002	(0.054)	0.015	(0.078)	-0.002	(0.054)
1	-0.071	(0.039)	-0.089	(0.064)	-0.041	(0.086)	-0.047	(0.055)
2	-0.112	(0.034)	-0.039	(0.053)	-0.088	(0.087)	0.033	(0.074)
3	-0.019	(0.029)	-0.091	(0.047)	-0.056	(0.099)	0.113	(0.077)
4	-0.044	(0.036)	0.051	(0.049)	-0.115	(0.090)	0.100	(0.071)
5	-0.067	(0.040)	-0.027	(0.060)	-0.117	(0.081)	0.143	(0.084)
6	-0.007	(0.032)	0.101	(0.046)	-0.059	(0.078)	0.213	(0.079)
7	0.019	(0.037)	0.041	(0.041)	-0.031	(0.064)	0.115	(0.078)
8	0.041	(0.037)	-0.006	(0.047)	-0.056	(0.095)	0.082	(0.071)
9	-0.134	(0.035)	-0.048	(0.057)	-0.266	(0.090)	-0.002	(0.067)
10	-0.226	(0.034)	-0.096	(0.053)	-0.280	(0.100)	-0.118	(0.050)
11	-0.305	(0.036)	-0.196	(0.047)	-0.382	(0.109)	-0.171	(0.058)
12	-0.201	(0.035)	-0.164	(0.052)	-0.342	(0.089)	-0.166	(0.075)
13	-0.204	(0.036)	-0.138	(0.054)	-0.318	(0.104)	-0.086	(0.061)
14	-0.196	(0.040)	-0.096	(0.049)	-0.248	(0.118)	0.008	(0.074)
15	-0.144	(0.038)	-0.054	(0.055)	-0.160	(0.119)	0.070	(0.083)
16	-0.187	(0.041)	-0.015	(0.061)	-0.223	(0.109)	-0.054	(0.079)
17	-0.144	(0.030)	-0.064	(0.051)	-0.206	(0.108)	0.059	(0.082)
18	-0.092	(0.030)	0.010	(0.044)	-0.124	(0.109)	0.007	(0.086)
19	-0.047	(0.040)	-0.062	(0.054)	-0.075	(0.097)	-0.010	(0.084)
20	-0.078	(0.038)	-0.156	(0.060)	-0.104	(0.096)	-0.029	(0.089)
21	-0.068	(0.041)	-0.049	(0.060)	0.001	(0.099)	-0.049	(0.070)
22	-0.085	(0.054)	-0.138	(0.062)	-0.050	(0.110)	-0.105	(0.076)
23	-0.106	(0.046)	-0.055	(0.057)	-0.038	(0.110)	-0.045	(0.072)
24	-0.051	(0.044)	-0.080	(0.058)	-0.053	(0.115)	-0.039	(0.075)
25	-0.064	(0.036)	-0.069	(0.069)	-0.056	(0.091)	-0.059	(0.078)

Notes: Standard-errors are clustered at the microregion level. See notes in Table 2.

Table A.7: Difference in differences, by maternal age group, Pernambuco versus Non-NE microregions.

months since emergency announced	Live births to mothers 24 and younger [1]		Live births to mothers 25 to 29 [2]		Live births to mothers 30 to 34 [3]		Live births to mothers 35 and older [4]	
-10	-0.054	(0.031)	0.009	(0.052)	-0.098	(0.044)	-0.013	(0.070)
-9	0.006	(0.028)	0.071	(0.043)	-0.033	(0.050)	-0.071	(0.066)
-8	0.039	(0.033)	0.003	(0.042)	0.017	(0.037)	0.007	(0.057)
-7	-0.051	(0.031)	0.023	(0.044)	-0.007	(0.050)	0.004	(0.066)
-6	-0.031	(0.033)	0.024	(0.040)	0.037	(0.046)	-0.019	(0.059)
-5	0.014	(0.031)	0.006	(0.043)	-0.039	(0.062)	0.017	(0.047)
-4	0.053	(0.030)	0.023	(0.044)	-0.030	(0.041)	0.047	(0.061)
-3	-0.063	(0.039)	0.056	(0.041)	0.058	(0.054)	-0.128	(0.065)
-2	0.004	(0.029)	0.044	(0.044)	0.112	(0.041)	0.000	(0.056)
-1	-0.033	(0.035)	-0.029	(0.050)	-0.021	(0.051)	-0.062	(0.086)
0	-0.016	(0.028)	-0.045	(0.046)	0.030	(0.059)	-0.025	(0.064)
1	-0.062	(0.028)	-0.098	(0.054)	-0.087	(0.059)	-0.026	(0.078)
2	-0.057	(0.031)	-0.014	(0.045)	-0.101	(0.047)	-0.176	(0.066)
3	-0.060	(0.027)	-0.029	(0.047)	0.011	(0.049)	-0.038	(0.063)
4	-0.011	(0.021)	-0.050	(0.041)	-0.023	(0.054)	0.120	(0.075)
5	-0.047	(0.032)	-0.054	(0.040)	-0.010	(0.048)	-0.063	(0.069)
6	0.034	(0.029)	0.053	(0.043)	0.100	(0.045)	0.074	(0.073)
7	0.023	(0.034)	0.000	(0.042)	0.013	(0.050)	0.094	(0.061)
8	0.036	(0.036)	0.034	(0.041)	-0.007	(0.049)	-0.030	(0.062)
9	-0.087	(0.034)	-0.042	(0.043)	-0.187	(0.053)	-0.229	(0.058)
10	-0.095	(0.030)	-0.224	(0.045)	-0.247	(0.057)	-0.322	(0.080)
11	-0.183	(0.034)	-0.312	(0.041)	-0.337	(0.063)	-0.273	(0.068)
12	-0.099	(0.031)	-0.238	(0.047)	-0.239	(0.059)	-0.315	(0.072)
13	-0.112	(0.030)	-0.217	(0.054)	-0.231	(0.052)	-0.222	(0.064)
14	-0.057	(0.034)	-0.187	(0.051)	-0.149	(0.053)	-0.143	(0.070)
15	-0.073	(0.033)	-0.110	(0.043)	-0.072	(0.061)	-0.228	(0.076)
16	-0.081	(0.030)	-0.154	(0.042)	-0.135	(0.041)	-0.265	(0.156)
17	-0.080	(0.023)	-0.095	(0.041)	-0.161	(0.054)	-0.201	(0.069)
18	-0.008	(0.028)	-0.040	(0.040)	0.004	(0.052)	-0.182	(0.064)
19	-0.038	(0.034)	-0.028	(0.031)	-0.049	(0.049)	-0.076	(0.073)
20	-0.079	(0.034)	-0.053	(0.039)	-0.147	(0.064)	-0.139	(0.063)
21	-0.050	(0.038)	-0.029	(0.044)	0.000	(0.049)	-0.150	(0.054)
22	-0.085	(0.039)	-0.091	(0.045)	-0.021	(0.062)	-0.122	(0.076)
23	-0.068	(0.028)	-0.092	(0.036)	0.013	(0.058)	-0.189	(0.070)
24	-0.040	(0.034)	-0.023	(0.060)	0.034	(0.059)	-0.113	(0.066)
25	-0.042	(0.030)	-0.067	(0.044)	-0.129	(0.055)	0.010	(0.071)

Notes: Standard-errors are clustered at the microregion level. See notes in Table 2.

Table A.8: Robustness of difference in differences, Pernambuco versus Non-NE microregions.

months since emergency announced	Original result [1]		Expanded time series [2]		Non Mosquito Infested NE [3]		Sample non-Infested NE [4]		Infested vs. non-Infested NE [5]	
-10	-0.051	(0.028)	-0.019	(0.028)	-0.055	(0.022)	-0.041	(0.022)	-0.041	(0.022)
-9	0.012	(0.027)	0.024	(0.027)	0.005	(0.026)	-0.007	(0.026)	0.020	(0.026)
-8	0.035	(0.030)	0.044	(0.027)	0.068	(0.031)	0.081	(0.031)	0.082	(0.030)
-7	-0.010	(0.028)	-0.002	(0.027)	0.008	(0.027)	0.009	(0.026)	0.022	(0.028)
-6	-0.013	(0.025)	-0.043	(0.024)	-0.011	(0.028)	-0.023	(0.028)	0.003	(0.028)
-5	0.008	(0.026)	0.001	(0.027)	0.011	(0.024)	-0.001	(0.024)	0.025	(0.026)
-4	0.030	(0.023)	0.042	(0.027)	0.045	(0.018)	0.031	(0.018)	0.059	(0.023)
-3	-0.034	(0.031)	-0.034	(0.023)	-0.015	(0.024)	-0.024	(0.024)	-0.001	(0.032)
-2	0.030	(0.028)	0.017	(0.022)	0.010	(0.025)	0.027	(0.025)	0.024	(0.028)
-1	-0.038	(0.033)	-0.027	(0.030)	-0.028	(0.031)	-0.031	(0.031)	-0.014	(0.031)
0	-0.026	(0.029)	-0.044	(0.026)	-0.073	(0.027)	-0.065	(0.026)	-0.059	(0.030)
1	-0.063	(0.034)	-0.069	(0.030)	-0.082	(0.034)	-0.074	(0.034)	-0.068	(0.033)
2	-0.071	(0.030)	-0.039	(0.026)	-0.075	(0.028)	-0.055	(0.028)	-0.051	(0.031)
3	-0.037	(0.026)	-0.025	(0.024)	-0.044	(0.024)	0.015	(0.021)	-0.020	(0.038)
4	-0.006	(0.028)	0.003	(0.022)	0.027	(0.028)	0.051	(0.027)	0.051	(0.041)
5	-0.052	(0.035)	-0.044	(0.030)	-0.035	(0.037)	-0.027	(0.037)	-0.011	(0.045)
6	0.045	(0.023)	0.015	(0.023)	0.047	(0.023)	0.052	(0.022)	0.071	(0.031)
7	0.028	(0.026)	0.021	(0.025)	0.031	(0.025)	0.039	(0.024)	0.055	(0.040)
8	0.022	(0.030)	0.035	(0.031)	0.037	(0.026)	0.016	(0.026)	0.061	(0.033)
9	-0.101	(0.029)	-0.101	(0.024)	-0.083	(0.022)	-0.078	(0.022)	-0.059	(0.031)
10	-0.175	(0.031)	-0.188	(0.034)	-0.196	(0.031)	-0.202	(0.031)	-0.172	(0.030)
11	-0.258	(0.032)	-0.246	(0.032)	-0.248	(0.031)	-0.267	(0.032)	-0.224	(0.036)
12	-0.176	(0.029)	-0.195	(0.031)	-0.223	(0.027)	-0.262	(0.028)	-0.199	(0.034)
13	-0.159	(0.026)	-0.165	(0.025)	-0.178	(0.026)	-0.221	(0.026)	-0.154	(0.033)
14	-0.129	(0.031)	-0.097	(0.028)	-0.134	(0.034)	-0.197	(0.035)	-0.100	(0.050)
15	-0.100	(0.028)	-0.088	(0.028)	-0.107	(0.027)	-0.154	(0.029)	-0.073	(0.048)
16	-0.122	(0.033)	-0.112	(0.027)	-0.089	(0.032)	-0.060	(0.032)	-0.055	(0.050)
17	-0.106	(0.023)	-0.098	(0.019)	-0.088	(0.022)	-0.109	(0.022)	-0.054	(0.046)
18	-0.040	(0.019)	-0.070	(0.024)	-0.038	(0.022)	-0.019	(0.020)	-0.005	(0.043)
19	-0.037	(0.031)	-0.044	(0.031)	-0.035	(0.031)	-0.005	(0.029)	-0.002	(0.049)
20	-0.095	(0.034)	-0.083	(0.031)	-0.081	(0.032)	-0.071	(0.031)	-0.047	(0.052)
21	-0.046	(0.034)	-0.046	(0.028)	-0.027	(0.028)	-0.006	(0.027)	0.006	(0.047)
22	-0.094	(0.035)	-0.107	(0.031)	-0.114	(0.030)	-0.114	(0.030)	-0.081	(0.051)
23	-0.074	(0.031)	-0.063	(0.024)	-0.064	(0.025)	-0.056	(0.024)	-0.031	(0.048)
24	-0.046	(0.035)	-0.065	(0.027)	-0.094	(0.031)	-0.070	(0.030)	-0.060	(0.050)
25	-0.046	(0.030)	-0.052	(0.024)	-0.065	(0.028)	-0.074	(0.028)	-0.031	(0.050)
Controls										
microregion	YES		YES		YES		YES		YES	
year*month	YES		YES		YES					
year							YES			
month							YES			
microregion*month	YES		YES							
microregion linear trends									YES	

Notes: Standard-errors are clustered at the microregion level. See notes in Table 2. Expanded time series includes observations for 2011 and 2012.

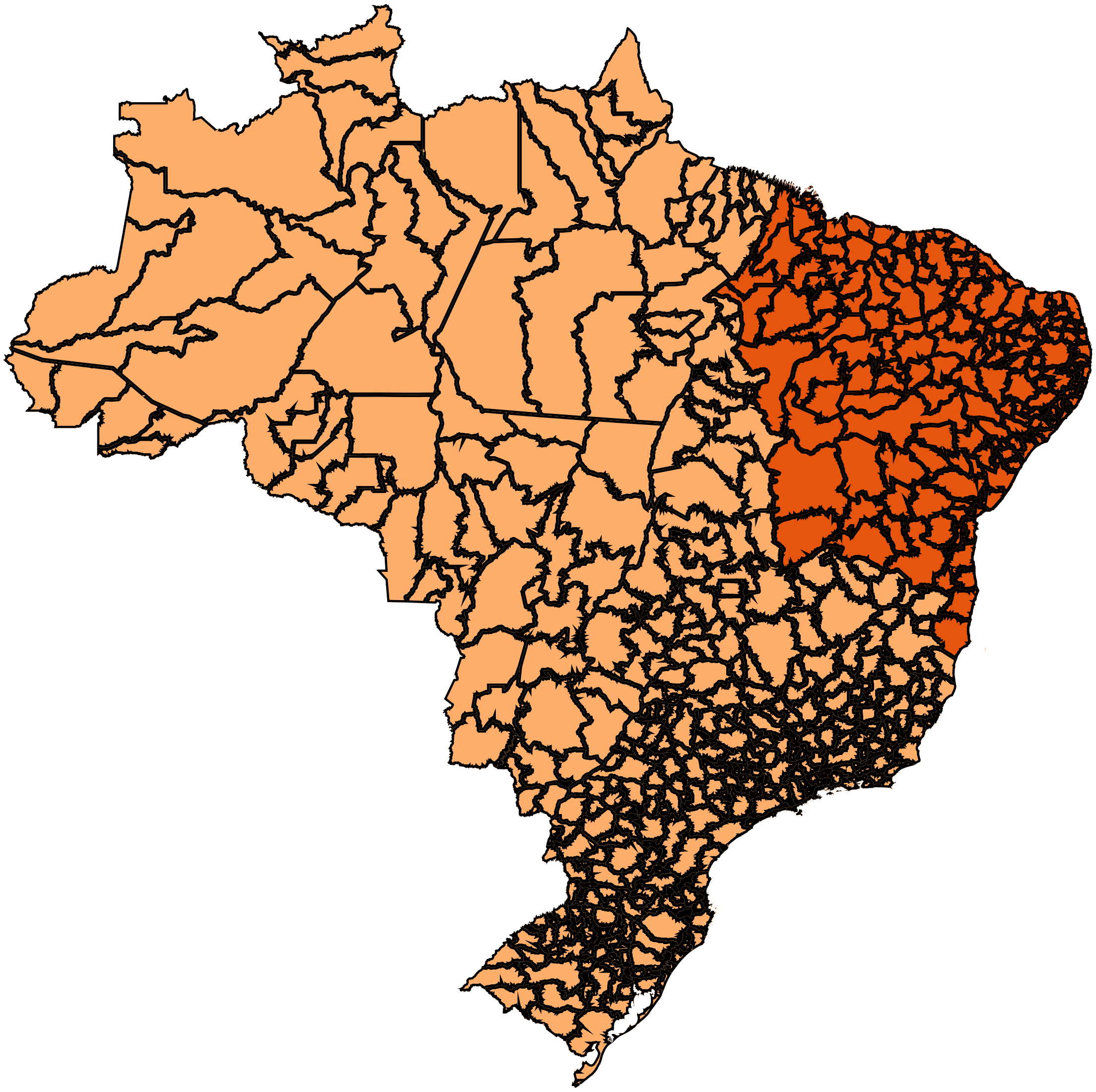

Figure A.1: Map of Brazil and its Microregions (*Regioes Geograficas Imediatas*)

Note: Northeast Region highlighted with darker shade.

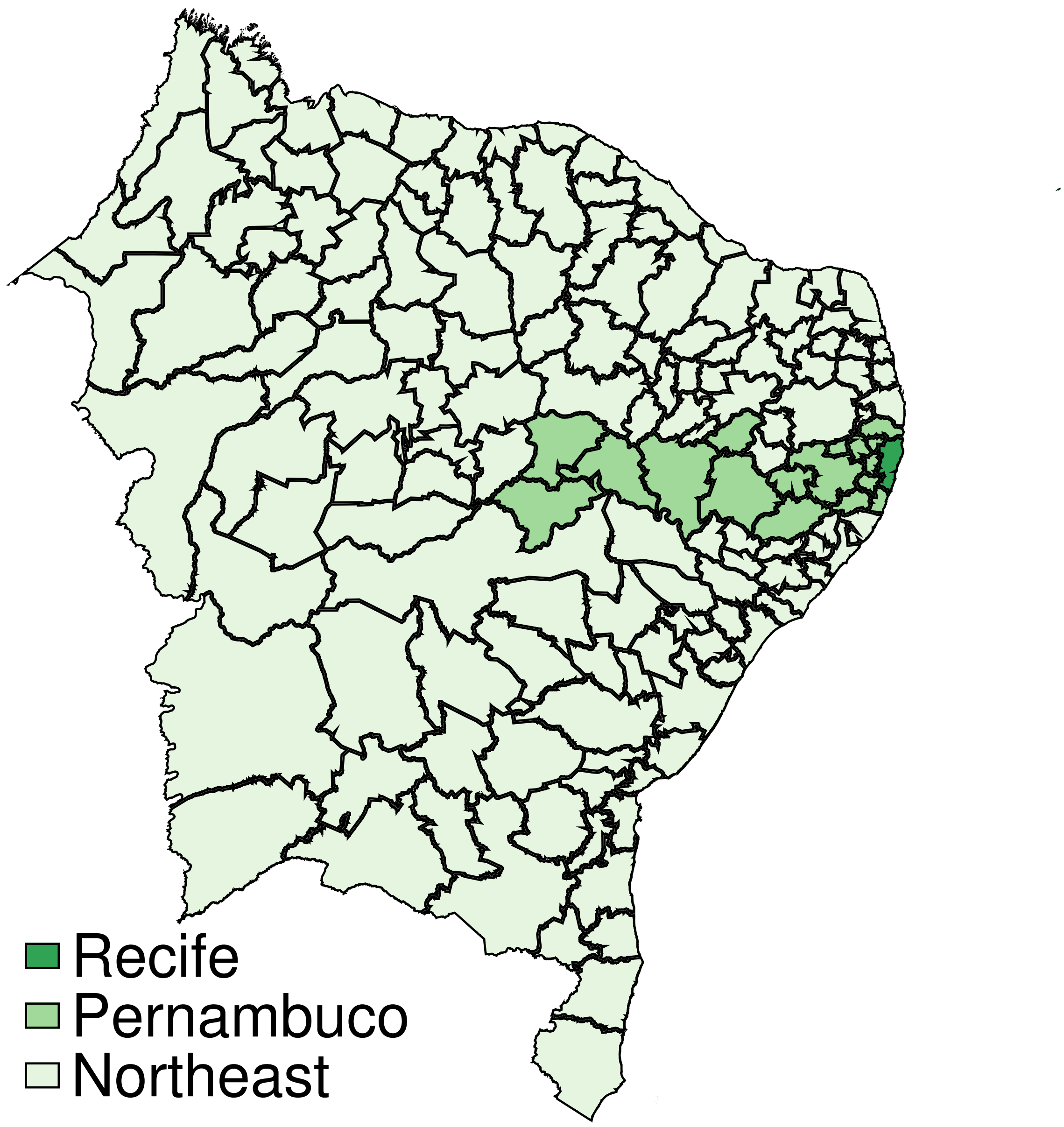

Figure A.2: Map of Brazilian Northeast and its Microregions (*Regioes Geograficas Imediatas*)

Note: Pernambuco state and Recife microregion highlighted with darker shades.